ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF COMMUNICATIONS AND MEDIA (RSCOM)

COURSE STRUCTURE & SYLLABUS

(BASED ON NATIONAL EDUCATION POLICY 2020)

For

B.Sc. IN ANIMATION AND VISUAL EFFECTS
(4 years Single Major)

W.E.F

AY - 2025 – 26

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Preamble

The National Education Policy (NEP) 2020 conceives a new vision for India's higher education system. It recognizes that higher education plays an extremely important role in promoting equity, human as well as societal well-being and in developing India as envisioned in its Constitution. It is desired that higher education will significantly contribute towards sustainable livelihoods and economic development of the nation as India moves towards becoming a knowledge economy and society.

If we focus on the 21st century requirements, the higher education framework of the nation must aim to develop good, thoughtful, well-rounded, and creative individuals and must enable an individual to study one or more specialized areas of interest at a deep level, and also develop character, ethical and Constitutional values, intellectual curiosity, scientific temper, creativity, spirit of service, and twenty-first-century capabilities across a range of disciplines including sciences, social sciences, arts, humanities, languages, as well as professional, technical, and vocational subjects. A higher quality education should be capable enough to enable personal accomplishment and enlightenment, constructive public engagement, and productive contribution to the society. Overall, it should focus on preparing students for more meaningful and satisfying lives and work roles and enable economic independence.

Towards the attainment of holistic and multidisciplinary education, the flexible curricula of the University will include credit-based courses, projects in the areas of community engagement and service, environmental education, and value-based education. As part of holistic education, students will also be provided with opportunities for internships with local industries, businesses, artists, crafts persons, and so on, as well as research internships with faculty and researchers at the University, so that students may actively engage with the practicalaspects of their learning and thereby improve their employability.

The undergraduate curriculums are diverse and have varied subjects to be covered to meet the needs of the programs. As per the recommendations of the UGC, introduction of courses related to Indian Knowledge System (IKS) is being incorporated in the curriculum structure which encompasses all of the systematized disciplines of Knowledge which were developed to a high degree of sophistication in India from ancient times and all of the traditions and practices that the various communities of India—including the tribal communities—have evolved, refined and preserved over generations, like for example Vedic Mathematics, Vedangas, Indian Astronomy, Fine Arts, Metallurgy, etc.

At RGU, we are committed that at the societal level, higher education will enable each student to develop themselves to be an enlightened, socially conscious, knowledgeable, and skilled citizen who can find and implement robust solutions to its own problems. For the students at the University, Higher education is expected to form the basis for knowledge creation and innovation thereby contributing to a more vibrant, socially engaged, cooperative community leading towards a happier, cohesive, cultured, productive, innovative, progressive, and prosperous nation."

Introduction

The National Education Policy (NEP) 2020 clearly indicates that higher education plays an extremely important role in promoting human as well as societal well-being in India. As envisioned in the 21st-century requirements, quality higher education must aim to develop good, thoughtful, well-rounded, and creative individuals. According to the new education policy, assessments of educational approaches in undergraduate education will integrate the humanities and arts with Science, Technology, Engineering and Mathematics (STEM) that will lead to positive learning outcomes. This will lead to develop creativity and innovation, critical thinking and higher-order thinking capacities, problem-solving abilities, teamwork, communication skills, more in-depth learning, and mastery of curricula across fields, increases in social and moral awareness, etc., besides general engagement and enjoyment of learning and more in-depth learning.

The NEP highlights that the following fundamental principles that have a direct bearing on the curricula would guide the education system at large, viz.

i. Recognizing, identifying, and fostering the unique capabilities of each student to promote her/his holistic development.
ii. Flexibility, so that learners can select their learning trajectories and programmes and thereby choose their own paths in life according to their talents and interests.
iii. Multidisciplinary and holistic education across the sciences, social sciences, arts,

humanities, and sports for a multidisciplinary world.

iv. Emphasis on conceptual understanding rather than rote learning, critical thinking to encourage logical decision-making and innovation; ethics and human & constitutional values, and life skills such as communication, teamwork, leadership, and resilience.

v. Extensive use of technology in teaching and learning, removing language barriers, increasing access for Divyang students, and educational planning and management.

vi. Respect for diversity and respect for the local context in all curricula, pedagogy, and policy.

vii. Equity and inclusion as the cornerstone of all educational decisions to ensure that all students can thrive in the education system and the institutional environment are responsive to differences to ensure that high-quality education is available for all.

viii. Rootedness and pride in India, and its rich, diverse, ancient, and modern culture, languages, knowledge systems, and traditions.

Approach to Curriculum Planning

Choice Based Credit System (CBCS) By UGC

Under the CBCS system, the requirement for awarding a degree or diploma or certificates prescribed in terms of number of credits to be earned by the students. This framework is being implemented in several universities across States in India. The main highlights of CBCS are as below [2]:

- The CBCS provides flexibility in designing curriculum and assigning credits based on the course content and learning hours.
- The CBCS provides for a system wherein students can take courses of their choice, learn at their own pace, undergo additional courses, and acquire more than the required credits, and adopt an interdisciplinary approach to learning.
- CBCS also provides an opportunity for vertical mobility to students from a bachelor's degree programmes to masters and research degree programmes.

Definitions

Academic Credit:

An academic credit is a unit by which a course is weighted. It is fixed by the number of hours of instructions offered per week. As per the National Credit Framework [2].

1 Credit = 30 NOTIONAL CREDIT HOURS (NCH)

Yearly Learning Hours = 1200 Notional Hours (40 Credits x 30 NCH)

30 Notional Credit Hours		
Lecture/Tutorial	**Practicum**	**Experiential Learning**
1 Credit = 15 -22 Lecture Hours	10-15 Practicum Hours	0-8 Experiential Learning Hours

Course of Study:

Course of study indicate pursuance of study in a particular discipline/programme. Discipline/Programmes shall offer Major Courses (Core), Minor Courses, Skill Enhancement Courses (SEC), Value Added Courses (VAC), Ability Enhancement Compulsory Courses (AECCs) and Interdisciplinary courses.

Disciplinary Major:

The major would provide the opportunity for a student to pursue in-depth study of a particular subject or discipline. Students may be allowed to change major within the broad discipline at the end of the second semester by giving her/him sufficient time to explore interdisciplinary courses during the first year. Advanced-level disciplinary/interdisciplinary courses, a course in research methodology, and a project/dissertation will be conducted in the seventh semester. The final semester will be devoted to seminar presentation, preparation, and submission of project report/dissertation. The project work/dissertation will be on a topic in the disciplinary programme of study or an interdisciplinary topic.

Disciplinary/interdisciplinary minors:

Students will have the option to choose courses from disciplinary/interdisciplinary minors and skill-based courses. Students who take a sufficient number of courses in a discipline or an interdisciplinary area of study other than the chosen major will qualify for a minor in that discipline or in the chosen interdisciplinary area of study. A student may declare the choice of the minor at the end of the second semester, after exploring various courses.

Courses from Other Disciplines (Interdisciplinary):

All UG students are required to undergo 3 introductory-level courses relating to any of the broad disciplines given below. These courses are intended to broaden the intellectual experience and form part of liberal arts and science education. Students are not allowed to choose or repeat courses already undergone at the higher secondary level (12^{th} class) in the proposed major and minor stream under this category.

Natural and Physical Sciences: Students can choose basic courses from disciplines such as Natural Science, for example, Biology, Botany, Zoology, Biotechnology, Biochemistry, Chemistry, Physics, Biophysics, Astronomy and Astrophysics, Earth and Environmental Sciences, etc.

i. ***Mathematics, Statistics, and Computer Applications:*** Courses under this category will facilitate the students to use and apply tools and techniques in their major and minor disciplines. The course may include training in programming software like Python among others and applications software like STATA, SPSS, Tally, etc. Basic courses under this category will be helpful for science and social science in data analysis and the application of quantitative tools.

ii. ***Library, Information, and Media Sciences:*** Courses from this category will help the students to understand the recent developments in information and media science (journalism, mass media, and communication)

iii. ***Commerce and Management:*** Courses include business management, accountancy, finance, financial institutions, fintech, etc.,

iv. ***Humanities and Social Sciences:*** The courses relating to Social Sciences, for example, Anthropology, Communication and Media, Economics, History, Linguistics, Political Science, Psychology, Social Work, Sociology, etc. will enable students to understand the individuals and their social behaviour, society, and nation. Students be introduced to survey methodology and available large-scale databases for India. The courses under humanities include, for example, Archaeology, History, Comparative Literature, Arts & Creative expressions, Creative Writing and Literature, language(s), Philosophy, etc., and interdisciplinary courses relating to humanities. The list of Courses can include interdisciplinary subjects such as Cognitive Science, Environmental Science, Gender Studies, Global Environment & Health, International Relations, Political Economy and Development, Sustainable Development, Women's, and Gender Studies, etc. will be useful to understand society.

Value-Added Courses (VAC):

Understanding India: The course aims at enabling the students to acquire and demonstrate the knowledge and understanding of contemporary India with its historical perspective, the basic framework of the goals and policies of national development, and the constitutional obligations with special emphasis on constitutional values and fundamental rights and duties. The course would also focus on developing an understanding among student-teachers of the Indian knowledge systems, the Indian education system, and the roles and obligations of teachers to the nation in general and to the school/community/society. The course will attempt to deepen knowledge about and understanding of India's freedom struggle and of the values and ideals that it represented to develop an appreciation of the contributions made by people of all sections and regions of the country, and help learners understand and cherish the values enshrined in the Indian Constitution and to prepare them for their roles and responsibilities as effective citizens of a democratic society.

i. Environmental science/education: The course seeks to equip students with the ability to apply the acquired knowledge, skills, attitudes, and values required to take appropriate actions for mitigating the effects of environmental degradation, climate change, and pollution, effective waste management, conservation of biological diversity, management of biological resources, forest and wildlife conservation, and sustainable development and living. The course will also deepen the knowledge and understanding of India's environment in its totality, its interactive processes, and its effects on the futurequality of people's lives.

ii. Digital and technological solutions: Courses in cutting-edge areas that are fast gaining prominences, such as Artificial Intelligence (AI), 3-D machining, big data analysis, machine learning, drone technologies, and Deep learning with important applications to health, environment, and sustainable living that will be woven into undergraduate education for enhancing the employability of the youth.

iii. Health & Wellness, Yoga education, sports, and fitness: Course components relating to health and wellness seek to promote an optimal state of physical, emotional, intellectual, social, spiritual, and environmental well-being of a person. Sports and fitness activities will be organized outside the regular institutional working hours. Yoga education would focus on preparing the students physically and mentally for the integration of their physical, mental, and spiritual faculties, and equipping them with basic knowledge about one's personality, maintaining self-discipline and self-control, to learn to handle oneself well in all life situations. The focus of sports and fitness components of the courses will be on the improvement of physical fitness including the improvement of various components of physical and skills-related fitness like strength, speed, coordination, endurance, and flexibility; acquisition of sports skills including motor skills as well as basic movement skills relevant to a particular sport; improvement of tactical abilities; and improvement of mental abilities.

These are a common pool of courses offered by different disciplines and aimed towards embedding ethical, cultural and constitutional values; promote critical thinking. Indian knowledge systems; scientific temperament of students.

Summer Internship /Apprenticeship:

The intention is induction into actual work situations**.** All students must undergo internships / Apprenticeships in a firm, industry, or organization or Training in labs with faculty and researchers in their own or other HEIs/research institutions during the ***summer term***. Students should take up opportunities for internships with local industry, business organizations, health and allied areas, local governments (such as panchayats, municipalities), Parliament or elected representatives, media organizations, artists, crafts persons, and a wide variety of organizations so that students may actively engage with the practical side of their learning and, as a by-product, further improve their employability. Students who wish to exit after the first two semesters will undergo 4-credit work-based learning/internship during the summer term to get a UG Certificate.

Community engagement and service: The curricular component of 'community engagement and service' seeks to expose students to the socio- economic issues in society so that the theoretical learnings can be supplemented by actual life experiences to generate solutions to real-life problems. This can be part of summer term activity or part of a major or minor course depending upon the major discipline.

Field-based learning/minor project: The field-based learning/minor project will attempt to provide opportunities for students to understand the different socio-economic contexts. It will aim at giving students exposure to development-related issues in rural and urban settings. It will provide opportunities for students to observe situations in rural and urban contexts, and to observe and study actual field situations regarding issues related to socioeconomic development. Students will be given opportunities to gain a first- hand understanding of the policies, regulations, organizational structures, processes, and programmes that guide the development process. They would have the opportunity to gain an understanding of the complex socio-economic problems in the community, and innovative practices required to generate solutions to the identified problems. This may be a summer term project or part of a major or minor course depending on study.

Indian Knowledge System:

In view of the importance accorded in the NEP 2020 to rooting our curricula and pedagogy in the Indian context all the students who are enrolled in the four-year UG programmes should be encouraged to take an adequate number of courses in IKS so that the ***total credits of the courses taken in IKS amount to at least five per cent of the total mandated credits*** *(i.e. min.8 credits for a 4 yr. UGP & 6 credits for a 3 yr. UGP)***.** The students may be encouraged to take these courses, preferably *during the first four semesters of the UG programme*. At least half of these mandated credits should be in courses in disciplines which are part of IKS and are related to the major field of specialization that the student is pursuing in the UG programme. They will be included as a part of the total mandated credits that the student is expected to take in the major field of specialization. The rest of the mandated credits in IKS can be included as a part of the mandated Multidisciplinary courses that are to be taken by every student. All the students should take a Foundational Course in Indian Knowledge System, which is designed to present an overall introduction to all the streams of IKS relevant to the UG programme. The foundational IKS course should be broad-based and cover introductory material on all aspects. Wherever possible, the students may be encouraged to choose a suitable topic related to IKS for their project work in the 7/8th semesters of the UG programme. [5]

(Note: Refer "Guidelines for Incorporating Indian Knowledge in Higher Education Curricula", University Grants Commission, March 2023 for further details)

Experiential Learning:

One of the most unique, practical & beneficial features of the National Credit Framework is assignment of credits/credit points/ weightage to the experiential learning including relevant experience and professional levels acquired/ proficiency/ professional levels of a learner/student. Experiential learning is of two types:

a. ***Experiential learning as part of the curricular structure*** of academic or vocational program. E.g., projects/OJT/internship/industrial attachments etc. This could be either within the Program- internship/ summer project undertaken relevant to the program being studied or as a part time employment (not relevant to the program being studied- up to certain NSQF level only). In case where experiential learning is a part of the curricular structure the credits would be calculated and assigned as per basic principles of NCrF i.e., 40 credits for 1200 hours of notional learning.

b. ***Experiential learning as active employment*** (both wage and self) post completion of an academic or vocational program. This means that the experience attained by a person after undergoing a particular educational program shall be considered for assignment of credits. This could be either Full or Part time employment after undertaking an academic/ Vocation program.

In case where experiential learning is as a part of employment the learner would earn credits as weightage. The maximum credit points earned in this case shall be double of the credit points earned with respect to the qualification/ course completed. The credit earned and assigned by virtue of relevant experience would enable learners to progress in their career through the work hours put in during a job/employment

Award of Degree in Animation and Visual Effects

The structure and duration of undergraduate programmes of study offered by the University as per NEP 2020 include:

Undergraduate programmes of either 3 or 4-year duration with Single Major, with multiple entry and exit options, with appropriate certifications:

UG Certificate: Students who opt to exit after completion of the first year and have secured 40 credits will be awarded a UG certificate if, in addition, they complete one vocational course of 4 credits during the summer vacation of the first year. These students are allowed to re-enter the degree programme within three years and complete the degree programme within the stipulated maximum period of seven years.

UG Diploma: Students who opt to exit after completion of the second year and have secured 80 credits will be awarded the UG diploma if, in addition, they complete one vocational course of 4 credits during the summer vacation of the second year. These students are allowed to re-enter within a period of three years and complete the degree programme within the maximum period of seven years.

3-year UG Degree: Students who will undergo a 3-year UG programme will be awarded UG Degree in the Major discipline after successful completion of three years, securing 120 credits and satisfying the minimum credit requirement.

4-year UG Degree (Honours): A four-year UG Honours degree in the major discipline will be awarded to those who complete a four-year degree programme with 160 credits and have satisfied the credit requirements as given in Table 6 in Section 5.

4-year UG Degree (Honours with Research): Students who secure 75% marks and above in the first six semesters and wish to undertake research at the undergraduate level can choose a research stream in the fourth year. They should do a research project or dissertation under the guidance of a Faculty Member of the University. The research project/dissertation will be in the major discipline. The students who secure 160 credits, including 12 credits from a research project/dissertation, will be awarded UG Degree (Honours with Research).

Award	Year	Credits to earn	Additional Credits	Re-entry allowed within (yrs)	Years to Complete
UG Certificate	1	40	4	3	7
UG Diploma	2	80	4	3	7
3-year UG Degree (Major)	3	120	x	x	x
4-year UG Degree (Honors)	4	160	x	x	X
4-year UG Degree (Honors with Research)	4	160	Students who secure cumulative 75%marks and above in the first six semesters		

Table: 1: Award of Degree and Credit Structure with ME-ME

Graduate Attributes

Sl.no.	Graduate Attribute	The Learning Outcomes Descriptors
GA1	Disciplinary Knowledge	A student will acquire knowledge and understanding of one or more disciplines. It will provide basic knowledge of Animation and Visual Effects use of creativity in CGI environment.
GA 2	Complex problem solving	The program focuses on good research and ability to identify solution-based thinking, application of theoretical concepts to real life case studies on Animation enabling students to develop problem solving skills.
		The students will be able to apply analytical thought including the analysis and evaluation of policies, and

GA 3	Analytical & Critical thinking	practices in the field of media and media relations. They will be able to identify relevant assumptions or implications. Identify logical flaws and holes in the arguments of others. Analyze and synthesize data from a variety of sources and draw valid conclusions and support them with evidence and examples.
GA 4	Creativity	A student will be able to draw connections between the knowledge gained and the creative task to be executed. Interpret the observations and sketch it into reality. A student will also be able Think 'out of the box' and generate solutions to complex problems in unfamiliar contexts by adopting innovative, imaginative, lateral thinking, interpersonal skills, and emotional intelligence.
GA 5	Communication Skills	A student will develop the ability listen carefully, read texts, and research papers analytically, and present complex information in a clear and concise manner to different groups/audiences
	Research-related skills	A Student will develop a keen sense of observation, inquiry, and capability for asking relevant/ appropriate questions. Should acquire the ability to problematize, synthesize and articulate issues and design research proposals, define problems, formulate appropriate and relevant research questions, formulate hypotheses, test hypotheses using quantitative and qualitative data, establish hypotheses, make inferences based on the analysis and interpretation of data, and predict cause-and-effect relationships. Should develop the ability to acquire the understanding of basic research ethics and skills in practicing/doing ethics in the field/ in personal research work.
GA 7	Collaboration	Capable of participating in project to work effectively and construct innovative end product in diverse teams both in classroom and in animation industry.
GA 8	Leadership readiness/qualities	A student will be able to operate and organize plan the tasks of a team or an organization and setting direction by formulating an inspiring vision and building a team that can help achieve the vision.
GA 9	Digital and technological skills	Demonstrate and experiment by other digital gadgets for learning, design, illustrate, and utilize relevant information using appropriate software for analysis of data and creation of product.
GA 10	Environmental awareness and action	A student will identify the effects of environmental degradation, climate change, and pollution. They will develop and illustrate the technique of spreading awareness on effective waste management, conservation of biological diversity, management of biological resources and biodiversity, forest, and wildlife

		conservation, and sustainable development and living by producing different Information Education and Communication (IEC) materials.

Program Learning Outcomes (PLO)

PLO-1: Acquiring Knowledge of Animation and Visual effects

A systematic or coherent understanding of the academic field of Animation, its different learning areas and applications, and its linkages with related disciplinary areas/subjects. Procedural knowledge that creates different types of professionals related to Animation & Visual effects area of study, including research and development, teaching and government and public service.

PLO-2: Ability of solving complex problem

The students attain ability to quickly identify the problem and applying critical thinking skills and problem-solving analysis in all dimensions of development and production

PLO-3 - Analytical & Critical thinking

The students will be able to apply analytical thought including the analysis and evaluation of policies, and practices in the field of media and media relations. Ability to understand and skills will be enhanced for identifying problems and issues relating to Animation and Visual effects

PLO-4: Develop and Demonstrate Creativity

A student will be able to demonstrate, perform, or think in different and diverse ways by using tools of new media about the objects and scenarios in the field of multimedia and deal with problems and situations that do not have simple solutions. They will be able to think 'out of the box' and generate solutions to complex problems in unfamiliar contexts by adopting innovative, imaginative, lateral thinking, interpersonal skills, and emotional intelligence.

PLO-5: Enhance and Execute Communication Skills

The students will develop the ability to listen carefully, read texts and research papers analytically, and present complex information in a clear and concise manner to different groups/audiences through various means of communication. A student will be able to express thoughts and ideas effectively in writing, through films and orally and communicate with others using appropriate media technologies.

PLO-6: Formulate Research-related skills

A Student will develop a keen sense of observation, inquiry, and capability for asking relevant/ appropriate questions. Should acquire the ability to problematize, synthesize and articulate issues and design research proposals, define problems, formulate appropriate and relevant research questions, formulate hypotheses, test hypotheses using quantitative and qualitative data, establish hypotheses, make inferences based on the analysis and interpretation of data, and predict cause-and-effect relationships. Students will develop the ability to acquire the understanding of basic research ethics and skills in

practicing/doing
ethics in the field/ in personal research work.

PLO-7: Collaboration

Capable to work effectively and respectfully with diverse teams in the classroom and in the media industry in the interests of a common cause and work efficiently as a member of a team.

PLO-8: Develop Leadership readiness/qualities

A student will be able to organize and operate the tasks of a team or an organization and setting direction by formulating an inspiring vision and building a team that can help achieve the vision.

PLO-9: Execute Digital and technological skills

The student will outline and examine using computers and other digital devices for learning, design, illustrate and utilize relevant information by using appropriate software is for analyzing of data and generate media related projects.

PLO 10: Identifying environmental issues, its awareness and action

A student will identify the effects of environmental degradation, climate change, and pollution. They will develop the technique and illustrate awareness on effective waste management, conservation of biological diversity, management of biological resources and biodiversity, forest and wildlife conservation, and sustainable development and living by producing different Information Education and Communication (IEC) materials.

Programme Specific Outcomes

PSO1: Integration of the concept, principles, and theories involved in the subject of animation and visual effects in all aspects

PSO2: Ability to identify and solve complex societal problems using different new media platforms.

PSO3: Student will be able to use their analytical thought in understanding different policies related to new media and its relationship in society.

PSO4: The student will be able to demonstrate ‘out of the box’ ideas by adopting innovative, imaginative, communicative skills and emotional intelligence.

PSO5: Ability to prepare, compare, and present complex information in a clear and concise manner to audience through various effective communication skills.

PSO6: Student will acquire the ability to identify and analyze societal related issues and design research proposals, understanding research & new media ethics, establish hypotheses and predict cause-and-effect relationships.

PSO7: Student will be skillful to connect and work effectively with diverse team of new media in the dynamic new media industry.

PSO8: Working effectively with different team members, student will devise and develop a leadership quality that can help them to achieve the vision in life.

PSO9: Student will demonstrate skills related to various digital devices, computers and appropriate software for analyzing data and new media related projects.

PSO10: Student will develop techniques and illustrate environmental awareness by producing different Information Education and Communication (IEC) materials.

Teaching Learning Process

Teaching and learning in this programme involve classroom lectures as well as tutorial and remedial classes.

Tutorial classes: Tutorials allow closer interaction between students and teacher as each student gets individual attention. The tutorials are conducted for students who are unable to achieve average grades in their weekly assessments. Tutorials are divided into three categories, viz. discussion-based tutorials (focusing on deeper exploration of course content through discussions and debates), problem-solving tutorials (focusing on problem solving processes and quantitative reasoning), and Q&A tutorials (students ask questions about course content and assignments and consolidate their learning in the guiding presence of the tutor).

Flip classroom: flip classroom allows lecture content from face-to-face class time to before class by assigning it as homework. This allows for more interactive forms of learning to take place during class

Experiential Learning: Experiential learning is a part of the academic curricular of the Animation and visual effects program. Following methods are adopted for Experiential Learning:

- Participating in intra and inter University competition, clubs of the university
- Extra-curricular activities like cultural activities, community outreach programmes,
- Projects and Portfolio
- OJT
- Internship
- Workshop and interacting with expert from the field
- Field trip, excursions, study tour, etc.
- Seminars and conferences organized in the department and University

Remedial classes: The remedial classes are conducted for students who achieve average and above average grades in their weekly assessments. The focus is laid to equip the students to perform better in the exams/assessments. The students are divided into small groups to provide dedicated learning support. Tutors are assigned to provide extra time and resources to help them understand concepts with advanced nuances. Small groups allow tutors to address their specific needs and monitor them. Following methods are adopted for tutorial and remedial classes:

- Written assignments and projects submitted by students
- Project or assignment -based learning
- Group discussions
- Home assignments
- Class tests, quizzes, debates organized in the department

Assessment Methods

<table>
<tr><th></th><th>Component of Evaluation</th><th>Marks</th><th>Frequency</th><th>Cod
e</th><th>Weightage
(%)</th></tr>
<tr><td>A</td><td>Continuous Evaluation</td><td></td><td></td><td></td><td></td></tr>
<tr><td>i</td><td>Analysis/Class test</td><td rowspan="5">Combination ofany three from
(i) to (v) with 5marks each</td><td>1-3</td><td>C</td><td rowspan="6">25%</td></tr>
<tr><td>ii</td><td>Home Assignment</td><td>1-3</td><td>H</td></tr>
<tr><td>iii</td><td>Project</td><td>1</td><td>P</td></tr>
<tr><td>iv</td><td>Seminar</td><td>1-2</td><td>S</td></tr>
<tr><td>v</td><td>Viva /Presentation</td><td>1-2</td><td>V</td></tr>
<tr><td>vi</td><td>MSE</td><td>MSE shall be of 10 marks</td><td>1-3</td><td>Q/C
T</td></tr>
<tr><td>vii</td><td>Attendance</td><td>Attendance shall be of 5 marks</td><td>100%</td><td>A</td><td>5%</td></tr>
<tr><td>B</td><td>Semester End Examination</td><td></td><td>1</td><td>SEE</td><td>70%</td></tr>
<tr><td></td><td>Project</td><td></td><td></td><td></td><td>100%</td></tr>
</table>

PROGRAME STRUCTURE

RSCOM
B.Sc. AVE

1st Semester

Sl. No.	Subject Code	Names of subjects	Course Level	Credit
		Major Papers		
1	AVE092M101	History of Animation and Multimedia	100	3
2	AVE092M112	Fundamentals of Drawing for Animation	100	3
		Minor Papers		
3	AVE092N113	Color Theory & Abstract	100	3
		Skill Enhancement Courses (SEC-1)		
4	AVE092S111	Characters & Illustration	100	3
		Value Added Course (VAC-1)		
5		Choose from the basket course	100	3
		Interdisciplinary Course (IDC-1)		
6		Indian Knowledge System I	100	3
		Ability Enhancement Course (AEC-1)		
7	AEC982A101	Communicative English and Behavioral Science-I	100	2
		Total -		**20**

2nd Semester

Sl. No.	Subject Code	Names of subjects	Course Level	Credit
		Major Papers		
1	AVE092M211	Concept Art and Digital Painting	100	3
2	AVE092M212	3D Modelling and Texturing	100	3
		Minor Papers		
3	AVE092N211	Graphic Design	100	3
		Skill Enhancement Courses (SEC-2)		
4	AVE092S211	Introduction to Cinematography	100	3
		Value Added Course (VAC-2)		
5		Choose from the basket course	100	3
		Interdisciplinary Course		
6		Indian Knowledge System II	100	3
		Ability Enhancement Course (AEC-2)		
7	AEC982A201	Communicative English and Behavioural Science-II	100	2
		Total -		**20**
Conferring the Certificate in Animation & Visual Effects, (CAVE)				

3rd Semester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M311	2D Animation	200	4
2	AVE092M312	3D Lighting and Rendering	200	4
Minor Papers				
3	AVE092N311	Introduction to 3d	200	4
Skill Enhancement Courses (SEC-3)				
4	AVE092S313	Introduction to Visual Effects	200	3
Interdisciplinary Course				
5		Choose from the basket course	200	3
Ability Enhancement Course (AEC-3)				
6		Communicative English-III	200	1
7		Behavioral Science-III	200	1
		Total -		20
4th Semester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M411	2D Animation FX and Compositing	200	4
2	AVE092M412	Rigging & 3D Animation Techniques	200	4
3	AVE092M413	Introduction to Chitra Art with Mo-graph (IKS based)	200	4
Minor Papers				
4	AVE092N411	Basics of Video Editing	200	3
5	AVE092N412	Foundation of Visual Effects	200	3
Interdisciplinary Course				
6		Choose from the basket course	0	0
Ability Enhancement Course (AEC)				
7	AEC982A401	Communicative English & Behavioral Science-IV		2
		Total -		20
5th Semester				
Sl. No.	**Subject Code**	**Names of subjects**	**Course Level**	**Credit**
Major Core Papers				
1	AVE092M511	Compositing & CGI Integration	300	4
2	AVE092M512	Overview of Dynamics	300	4
3	AVE092D511	DSE-1 Rigging and 2D Animation	300	4
Minor Papers				
4	AVE092N511	Stop Motion	300	4

5	AVE092I511	Internship	300	4
		Total -		20

6th Semester

Sl. No.	Subject Code	Names of subjects	Course Level	Credit
Major Core Papers				
1	AVE092M611	Post-production for 2d Animation	300	4
2	AVE092M612	Advanced 3D Dynamics	300	4
Discipline Specific Elective (DSE)				
3	AVE092D611	DSE-1 Procedural Animation and Compositing for Film	300	4
4	AVE092D602	DSE-2 Film Appreciation & Analysis	300	4
Minor Papers				
5	AVE092N611	Pencil to Pixel	300	4
		Total -		20

7th Semester

Sl. No.	Subject Code	Names of subjects	Course Level	Credit
Major Core Papers				
1	AVE092M701	Concept for Game Design and Prototype	400	4
2	AVE092M712	DSE I - Visual Effects Production	400	4
3	AVE092M703	Research Methodology	400	4
4	AVE092M714	Overview of UI/UX	400	4
Minor Papers (Other Dept)				
5	AVE092N711	Introduction to Architecture Modelling	400	4
		Total -		20

8th Semester

Sl. No.	SubjectCode	Names of subjects	Course Level	Credit
1	AVE092M701	Media Entrepreneurship and innovation	400	4
2	AVE092M712	Papercut Puppetry	400	4
3	AVE092M713	**Project Dissertation/Research Project** (Preferred Option)	400	12
4	AVE092M714	Short Film for 2D Animation	400	4
5	AVE092M715	3D Portfolio Development with Industry Pipeline	400	4
6	AVE092M716	VFX – Portfolio Development	400	4
		Total -		20

Semester I	
Major Paper-1	History of Animation and Multimedia
Subject Code	AVE092M101
Credit	3
LTPC	2-1-0-3

Course Objectives:

New interpretations of contemporary ideas of animation based on an understanding of history of animation.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify** the meaning, concept, and process of animation.	**BT 1**
CO 2	**Classify** the characteristic features of the different types of animation.	**BT 2**
CO 3	**Understand** the principles of animations theories in Multimedia.	**BT 2**
CO 4	**Interpret** the composition associated with the rise and evolution of animation.	**BT 2**

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods
I	**History of Animation** Influence of predecessors,1888–1909: Earliest animations on film, 1910s: From original artists to "assembly-line" production studios,1920s: Absolute film, synchronized sound and the rise of Disney,1930s: Color, depth, cartoon superstars and Snow White,1950s: Shift from classic theatrical cartoons to limited animation in TV series for children,2000s–2010s: traditional techniques overshadowed by computer animation.	15
II	**History of Mean Methods of Animation** Magic Lanterns and the zoetrope, Puppet Animation, Animated Series, Movies, History of comic and manga, Celluloid Animation, 2D Animation,3D Animation, Motion Graphics, Stop Motion, cut-out animation.	15
III	**History of Anime** Origins of Anime (early 1900-1922), Pre-war productions (1923-1939), During the Second World War, Post-war environment, Toei Animation and Mushi Production	15
IV	**History of Anime** Origins of Anime (early 1900-1922), Pre-war productions (1923-1939), During the Second World War, Post-war environment, Toei Animation and Mushi Production	15
	Total	**60**

Text Book:

- Blaise, Aaron. (2021). *The Art of Aaron Blaise.* Vol.1. ISBN 1737328801.
- Eugene, Emile. &Courted, Jean Louis. (1908). *Fantasmagorie.* Movie.

References:

- https://www.britannica.com/summary/animation
- http://animation-ua.com/en/school-animation/history-of-animation/178-history-of- animation

Credit Distribution		
Theory/Tutorial	Practicum	Experiential Learning
60 hrs.	NA	30 hrs. Workshop (10hrs), Presentation (preparation – 3hrs, presentation 40 min), Projects (3hrs 40 min), Case studies (6hrs 20min)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Two Workshop	NA	10	10
Presentations after observingthe workshop	3	40min (20 min each)	3hrs 40min
Projects	NA	10 hrs	10hrs
Case studies	NA	6hrs 20min	6hrs 20min
Total Hours			**30**

Semester I	
Major Paper-2	Fundamentals of Drawing for Animation
Subject Code	AVE092M112
Credit	3
LTPC	0-1-4-3

Course Objectives:

To equip students with knowledge of the foundational concepts of the art that will enable them tounderstand, draw different art style, study human and animal anatomy.

Course Outcomes:

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Relate to the** basics Skeleton, Muscle, Proportion of animal and human anatomy to portray the Art in the most compelling way.	**BT 1**
CLO 2	**Demonstrate the understanding of** shapes and forms, styles, andtraditions through familiarization with a wide range of design, styles, etc.	**BT 3**
CLO 3	**Executing** the knowledge of art in their attempts to draw prospective drawing and human figures.	**BT 3**
CLO 4	**Develop** new interpretations of contemporary ideas of design based on anunderstanding of Principles of Designs and Elements of Shape.	**BT 3**

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods
I	**Fundamentals of Animation** Principles of Designs and Elements of Shape, Shape Modification, Stick Figures, Gesture Drawing's Depth Study, Colour Theory; Doubt Clearing Assignments	**15**
II	**Human and Animal's Anatomy** Studies of Skeleton, Muscle, Proportion; Studies of Animal Skeleton, Muscles, Proportion; Self Hand and Feet Drawing; Animal Family Study, Head and Body Turn Around; Understanding the Form and Volume; Doubt Clearing Assignments	**15**
III	**Still life Using Perspectives** Exterior, Interior, Angle Design by Applying Linear Perspective; Still life Drawing and Composition, Pencil, Black and White and Colour, Rendering; Doubt Clearing Assignments	**15**
IV	**Character Designing Using Basic Shapes** Introduction to Character Drawing with Basic Shape, Expression Sheet, Character Style Study, Pose Board, Character Concept Art, Manual, Character Turn Around, Character Personality	**15**
TOTAL		**60**

Text Book:

- Drawing and Anatomy by Victor Perard (1928)
- Illusion of Life by Frank Thomas and Olli Johnston (1981)

References:

- https://www.youtube.com/watch?v=uDqjIdI4bF4
- https://www.youtube.com/watch?v=8J39SslgJsQ

Credit Distribution		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40hrs.	30 hrs. Live Sketching (15hrs), Photography for inspirationboard / reference (5hrs), Projects and Portfolio (10).

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Live Sketching (human)	NA	15	15
Photography for inspiration board / reference	NA	5	5
Projects for exhibition	NA	10	10
Total Hours			**30**

Semester I	
Minor Paper-1	Color Theory & Abstract
Subject Code	AVE092N113
Credit	3
LTPC	1-1-2-3

Course Objectives:

The objective of Colour Theory & Abstract is to enable the students to develop the knowledge of colour and its applications in different scenarios.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify and understand** the application and uses of colour	**BT 2**
CO 2	**Identify** the colour terminologies and theory.	**BT 3**
CO 3	**Demonstrate** and apply the different colour schemes on compositions	**BT 3**
CO 4	Be able to **analyze** colour psychology in real world scenarios	**BT 4**

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Abstract** Optical illusion, Pencil rendering, Value, composition	**15**
II	**Colour Wheel** Colour wheel – Primary, Secondary and Tertiary Colours	**15**
III	**Grey Scale** Whites & Blacks, Hues, Tints, and Shades	**15**
IV	**Colour Schemes** Monochromatic, Warm, Cool, Complimentary, Split Complimentary, Analogous, Triadic Colour	**15**
	TOTAL	**60**

Text Book:

- Interaction of Color by Josef Albers
- The Secret Lives of Color by Kassia St. Clair

References:

- The Colour Bible: The definitive guide to colour in art and design by Laura Perryman

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning

NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Extracurricular Activity	NA	10 hrs	10 hrs
Cultural activities	3 hrs	1 hrs	4 hrs
Street Play	4 hrs	1 hrs	5 hrs
Interacting with actor	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs
Total Hours			**30**

Semester I	
Skill Enhancement Courses -1	Color Theory & Abstract
Subject Code	AVE092S111
Credit	3
LTPC	1-0-4-3

Course Objectives:

A personal illustration style and technique that uses both traditional and digital skills and incorporatesacquired knowledge, experience, judgment, and unique aesthetic vision.

Course Outcomes

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Bloom's Taxonomy Level**
CO 1	**Classify** a character's traits, personality, backstory according to the story	**BT 1**
CO 2	**Summarize** different styles through which student can sketchdifferent themes or character design props.	**BT 2**
CO 3	**Experiment** the different techniques in illustration.	**BT 3**
CO 4	**Analyze** themes of different conceptual art of animation	**BT 4**

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Art Fundamentals** the Basic Structure of Art and Drawing, Breaking Downthe Structure, Skill vs. Emotion in Drawing, The Importance of Story in Drawing, Drawing Art Studies vs Drawing Art CreationDrawing Imaginative Art	**15**
II	**Character forms and proportion** Studies of Skeleton, Muscle, Proportion; Studies of Animal Skeleton, Muscles, Proportion; Self Hand and Feet Drawing; Animal Family Study, Head and Body Turn Around; Understanding the Form and Volume; Doubt ClearingAssignments	**15**
III	**Hair cloth Dynamics** Fundamental of drawing hair, Basic Components of drawing hair, drawing basic hair shapes, drawing clothing and cloth dynamics	**15**
IV	**Character post- production** Creating Clean Lines and Line Art for Finishing Drawings Creating Rough Clean Lines and Line Art for Finishing DrawingsCharacter Page Composition when Drawing Characters, Story board, Pose board, Inspiration board, Expression sheet	**15**
TOTAL		**60**

Text Book:

1. Creating Stylized Characters by Marisa Lewis
2. The Silver Way: Techniques, Tips, and Tutorials by Stephen Silver

References:

- https://www.youtube.com/watch?v=YytdLj89iXE
- https://www.youtube.com/watch?v=gI62rHNtg2w

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
20 hrs.	40 hrs.	30 hrs. Conceptual Sketching (10hrs),Project and Portfolio (15hrs), Photography for inspiration board / reference (2hrs), Case studies (3hrs)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Conceptual Sketching basedtrends	NA	10	10
Project and Portfolio basedtrends	NA	15	15
Photography for inspirationboard / reference	NA	2	2
Case studies	NA	3	3
Total Hours			**30**

Semester II	
Major Course -1	Concept Art and Digital Painting
Subject Code	AVE092M211
Credit	3
LTPC	1-0-4-3

Course Objectives:

To define the concepts, techniques, principles and practices in Advertising and Public Relations to classify the mysteries of media marketing, positioning, market segmentation and targeting in advertising as well as the significance of media in globalization.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Reproduce** rough concept of branding, art style, character's traits, personality, props according to the script and use that information as a guide for creating the visuals	**BT 1**
CO 2	**Visualize** different styles sketch different themes, cartoons, design, props and environment.	**BT 2**
CO 3	**Explore** themes of different conceptual art and design	**BT 3**
CO 4	**Construct** conceptual art and design using various tricks and technique	**BT 4**

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Workspace and Tools** Introduction to User Interfaces, Basic Setting, Text, Layer Management,Tools	**15**
I I	**Masking and Filters** Masking and Filters	**15**
I I I	**Digital Painting and Matte Painting** Still life Painting, how light falls on form, Dynamic light and shadow, Photo basing, digital painting technique and tips, Illustration, Manipulation,	**15**
IV	**Master Layout Design and Background** Master Layout Design, Rough Final layout(beats), Layout and Colouring	**15**

Text Book:

1. Bold Visions: A Digital Painting Bible
2. Digital Painting Techniques: Practical Techniques of Digital Art Masters” by 3D Total

References:

1. https://www.youtube.com/watch?v=fJvcllrOrr4
2. https://www.youtube.com/watch?v=XHprIlkY8Q4.

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Photography for inspiration board / reference/visualizationtechnique (6hrs), Project and Portfolio (15hrs), Conceptual Sketching (6hrs), Case studies(3hrs)

Break up of Experiential learning

Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Photography for inspiration board / reference/visualizationtechnique	NA	6	6
Project and Portfolio	NA	15	15
Conceptual Sketching	NA	6	6
Case studies	NA	3	3
Total Hours			**30**

Semester II	
Major Course -II	Introduction to 3D Modelling and Texturing
Subject Code	AVE092M212
Credit	3
LTPC	1-0-4-3

Course Objectives:

This course would likely enhance the student's abilities in both the technical aspects of 3D modeling and the creative aspects of design and texturing, allowing them to produce high-quality, visually appealing models for a variety of digital applications.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	Course Outcome	Blooms Taxonomy Level
CO 1	**Identifying** the object's shape, texture, and appearance using polygons, curves, and other geometric primitives.	BT 2
CO 2	**Construct** 3D model to view from any angle and use for a variety ofpurposes, such as visualization, animation, prototyping, etc.	BT 3
CO 3	**Apply** the knowledge of shapes and forms to model a character, object,and environment.	BT 3
CO 4	**Develop** new interpretations of contemporary ideas of 3d texturing	BT 4

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Modelling:** Introduction; Hardware/Software; Pipeline Demonstration; 3d Interface;Low Poly Modelling; Export/Import; Collision	15
II	**Surfacing:** High Poly Modelling; Normal Bake; Masking Height; Bake Other Maps; Substance Painter Materials; Export; Photoshop	15
III	**Modularity:** Wall Set – Model; Modularity: Wall Set – Unwrap; Modularity: Wall Set - Height & Normal; Modularity: Wall Set - Materials,Modularity: Wall Set - Export & Import to Unity; Modularity: Wall Set – Unity Prefabs; Modularity: Wall Set - Level Layout	15
IV	**Terrain:** Foliage: Palm Tree (model/unwrap), Foliage: Palm Tree, Terrain: SculptingHeight, Terrain: Adding Textures, Terrain: Adding Trees & Decorations; Terrain: Polish	15
	TOTAL	**60**

Text Book:

- Vaughan, William. (2011). Digital Modeling. Edition 1, ISBN 978-0321700896, New RidersPub
- Legaspi, Chris. (2015). Anatomy for 3D Artists: The Essential Guide for CG Professionals.3dtotal Publishing

References:

- https://www.youtube.com/watch?v=_UC0F3ZiuEU
- https://www.youtube.com/watch?v=YMzKgFed7vA

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Conceptual Prototype (10hrs), Project and Portfolio (15hrs), (3hrs), Case studies (5hrs)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Conceptual Prototype	NA	10	10
Project and Portfolio based ontrends	NA	15	15
Case studies	NA	5	5
		Total Hours	**30**

Semester II	
Minor Course -I	Graphic Design
Subject Code	AVE092N211
Credit	3
LTPC	1-1-2-3

Course Objectives:

To refine the techniques, principles, and practices in Branding to classify the mysteries of media marketing, positioning, market segmentation and targeting in advertising as well as the significance of media in globalization.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Compare** different design and themes	**BT 2**
CO 2	**Apply** the Graphic in difference type of Branding.	**BT 3**
CO 3	**Illustrate** ranges of images using different design techniques.	**BT 3**
CO 4	**Demonstrate** visual research and development skills through the creation of a Brand Development Guide	**BT 3**

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Fundamentals of Graphic Design** Introduction to interface of software, Implement the fundamentals of color, visual, rhythm, and pattern in design. Use scale, weight, direction, texture, and space in a composition Typeset text and experiment with letter forms Create your own series of images using different image makingtechniques	**15**
II	**Typography** Review the terminology and measuring system used to describe typeExplore how typefaces tell stories and understand the historic evolution Conduct a peer-reviewed typesetting exerciseDesign of a full-scale typographic poster	**15**
III	**Image Making** Make informed design choices using image-based research Create ranges of representation using images Compose spreads for book Design a book with your own images.	**15**
IV	**Branding** Synthesize typography, image making, composition and systematicthinking skills through ideation, invention, and conceptualization Demonstrate visual research and development skills through the creation of a Brand Development Guide Expand a brand identity's palette through the inclusion of graphicmarks or icons, color, secondary typefaces, and/or images	**15**
TOTAL		**60**

Text Book:

- **Grid Systems in Graphic Design**, by Josef Müller-Brockman
- **Paula Scher: Works**, by Tony Brook & Adrian Shaughnessy

References:

- https://www.youtube.com/watch?v=xntKz5gLoNI
- https://www.tourboxtech.com/en/news/best-graphic-design-books-for-designer.html

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs., 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	30 hrs. Project and Portfolio (15hrs), Study tour (5hrs), Photography for inspiration board / reference (5hrs), Casestudies (5hrs)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time(hrs.)
Project and Portfolio basedtrends	NA	15	15
Study Tour	NA	5	5
Photography for inspirationboard / reference	NA	5	5
Case studies	NA	5	5
Total Hours			**30**

Semester II	
Skill Enhancement Course II	Introduction to Cinematography
Subject Code	AVE092S211
Credit	3
LTPC	1-0-4-3

Course Objectives:

This course aims to provide a comprehensive understanding of the fundamental principles of filmmaking, with a focus on the use of cameras, lighting, scripting, character design, and storyboarding. It will empower students to harness these concepts to produce high-quality audio-visual content by the end of this course.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Understanding** the basics of the camera, it is working, and its functions and distinguish between various camera body parts, different shot sizes and various camera setups. Visualize the Rule of Thirds for better composition.	**BT 2**
CLO 2	**Relate** the fundamentals of lighting that affects the visual mood and atmosphere of a scene. It also helps with better storytelling, shaping character experience and enhancing the cinematic lighting techniques.	**BT 2**
CLO 3	**Construct** a script, design characters and story boarding.	**BT 3**
CLO 4	**Apply** the knowledge of camera, lighting, scripting, designing characters and story boarding to explore the new dimensions of audio video editing and to generate a final product.	**BT 3**

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Fundamentals of Camera:** Basics of Camera, its functions, Single camera and multi-camera setup, accessories, composition, Rule of thirds, different types of shots.	**15**
II	**Fundamentals of Lighting:** Importance of lighting,3 point Lighting, 5-point Lighting, Silhouette,	**15**
III	**Scripting:** Scripting writing, Character designing, Storyboarding, Animatic,	**15**
IV	**Editing:** Understanding interface and editing concept, uses of transition, effects and tools, audio integration, Exporting final output.	**15**
TOTAL		**60**

Textbook:

1. Brown, Blain; Cinematography: Theory and Practice, Second Edition: Image Making for Cinematographers and Directors; Focal Press, 2011.

2. Katz, D Steven; Film Directing Shot by Shot: Visualizing from Concept to Screen; MichaelWiese

References:

1. https://www.youtube.com/watch?v=KRC-6fMbw7k
2. https://www.youtube.com/watch?v=bVlTlTRhRDc

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs., 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	30 hrs. Project (5hrs), Study tour(4hrs), shooting (20hrs),

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Project	NA	5	5
Study Tour	NA	4	4
Shooting	NA	20	5
Interaction with industryexpert	NA	1	1
		Total Hours	**30**

Semester III	
Major Course -I	2D Animation
Subject Code	AVE092M311
Credit	4
LTPC	1-1-4-4

Course Objectives:

This course aims to provide students with a thorough understanding of the principles and techniques required for creating basics of animation. It will cover the process of reviewing storyboards and animatics, converting them into fully-fledged 2D animations, and applying key animation and design principles, such as color, shapes, biomechanics, and fluidity. By the end of the course, students will be able to create biomechanics of animated sequences that effectively communicate a narrative while maintaining visual consistency.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Review** storyboard and animatic sequence and overall flow of the story and whether the images effectively convey the intended narrative.	**BT 1**
CLO 2	**Convert** animatic sequence 2-dimensional images drawing or creating each individual frame of the animation by hand or using digital software	**BT 2**
CLO 3	**Apply** the knowledge of color, shapes, and forms to retain mood the mass of subject to create smooth sequence.	**BT 3**
CLO 4	**Demonstrate** the knowledge of Bio-Mechanics and **Construct** animated sequence	**BT 4**

Detailed Syllabus:

Modules	Course Contents	Periods
I	**2D Digital Animation:** Interface, Timeline, Properties and Library, Character Tracing, Sack animation, Ball animation and 12 principles of animation	15
II	**Bio-Mechanics / Organic Animation (Digital):** Head Turn, Walk, Jump (all view), Run Cycle, Walk-Run-Stop, Character balance, Lip chart	15
III	**Biomechanics of 2d Animation:** Fighting Scene, turn around, secondary animation, follow through animation. In depth study on Time and space	15
IV	**Animatic:** Video reel with dialogue and Time	15
TOTAL		**60**

Textbook:

- Williams, Richard. (2001). *The Animator's Survival Kit*
- *Timing for Animation, 40th Anniversary Edition (greyscale)*

References:

- https://www.youtube.com/watch?v=fJvcllrOrr4
- https://www.youtube.com/watch?v=XHprIlkY8Q4.

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Inspiration board / reference/visualization technique (6hrs), Project and Portfolio (15hrs), Conceptual Sketching (6hrs), Case studies (3hrs)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
inspiration board / reference/visualization technique	NA	6	6
Project and Portfolio	NA	15	15
Conceptual Sketching	NA	6	6
Case studies	NA	3	3
Total Hours			**30**

Semester III	
Major Course -I	3D Lighting and Rendering
Subject Code	AVE092M312
Credit	4
LTPC	1-0-4-3

Course Objectives:

Students will have a strong grasp of natural light properties, lighting nodes, and various rendering engines, as well as the skills to produce high-quality rendered images and walkthroughs. Students will also complete a final project to demonstrate their technical and creative abilities in 3D lighting and rendering.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Remembering** Key Concepts of Natural Light and basic lighting tools and whether the images effectively convey the intended narrative.	**BT 1**
CLO 2	**Relate** light node with the attributes	**BT 2**
CLO 3	**Apply** the features of different render engines by optimizing render settings for efficiency and quality	**BT 3**
CLO 4	**Analyze** new scenes by rendering in different type of lights.	**BT 4**

Detailed Syllabus:

Modules	Course Contents	Periods
I	**3D lighting – 1:** Introduction to nature light and its properties, basic lights of the software.	15
II	**3D lighting – 2:** Light nodes and its attributes.	15
III	**Rendering:** Arnold Engine, V-ray Engine, Corona Rendering, / Render-man Engine by Pixar Studios, Cycle render	15
IV	**Project**: Students will have to individually submit rendered images/walkthrough submit in a storage device. Teachers will supervise the projects.	15
TOTAL		**60**

Textbook:

- (2000). Digital Lighting and Rendering by **Birn Jeremy**.
- Lighting for Animation: The Art of Visual Storytelling" by **Jeanette Barnes**

References:

- https://www.youtube.com/watch?v=iUfCRXbL9ZM
- https://www.youtube.com/watch?v=kADhc51BkbI

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs., 3x30=90)		
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Conceptual Prototype (10hrs),Project and Portfolio (15hrs), (3hrs), Case studies (5hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Conceptual Prototype	NA	8	12
Project and Portfolio based on trends	NA	10	20
Case studies	NA	5	5
Total Hours			**30**

Semester III	
Minor Course -I	Introduction to 3D
Subject Code	AVE092N311
Credit	4
LTPC	0-1-4-3

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Identifying** the object's shape, texture, and appearance through the used basic primitives.	**BT 1**
CLO 2	**Construct** texturing for a variety of purposes, such as gaming props, animation etc.	**BT 2**
CLO 3	**Apply** the knowledge of lights to light a subject & object.	**BT 3**
CLO 4	**Analyze** new ideas by rendering different subjects & object.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Modelling** Introduction; Hardware/Software; Pipeline Demonstration; 3d Interface; Low Poly Modelling; Export/Import;	15
II	**Texturing** UV Unwrapping, Nodes, baking of texturing maps	15
III	**Lighting:** One-, Two- and Three-point lighting, Studio lighting, using HDRI Maps for Lighting.	15

IV	**Rendering** Render the object & subject using different renderer.	**15**
TOTAL		**60**

Text Book:

1. Vaughan, William. (2011). Digital Modeling. Edition 1, ISBN 978-0321700896, New Riders Pub
2. Legaspi, Chris. (2015). Anatomy for 3D Artists: The Essential Guide for CG Professionals. 3dtotal Publishing

References:

1. **https://www.youtube.com/watch?v=_UC0F3ZiuEU**
2. **https://www.youtube.com/watch?v=YMzKgFed7vA**

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Conceptual Prototype (10hrs), Project and Portfolio (15hrs), (3hrs), Case studies (5hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Conceptual Prototype	NA	10	10
Project and Portfolio based on trends	NA	15	15
Case studies	NA	5	5
		Total Hours	**30**

Skill Enhancement Courses:

Title of the Paper: Introduction to Visual Effects **Course Level: 100**

Subject Code: AVE092D313

L-T-P-D : 1-0-4-3 **Total credits: 3**

Course Objectives:

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Understand** and holographic *sequence* and overall flow the story and whether the images effectively convey the intended narrative.	**BT 1**
CLO 2	**Convert** live action world *Into Imaginary world* using VFX software	**BT 2**
CLO 3	**Apply** the knowledge of. VFX for creating FX in shorts	**BT 3**
CLO 4	**Construct** appealing *sequence using* VFX technique and tools	**BT 4**

COURSE OUTLINE:

Modules	**Course Contents**	**Periods**
I	**Introduction to After Effects** Introduction to the UI of software, composition, tools, encoder, and rendering, Keyframe Animation	**15**
II	**Introduction Motion Graphics** Creating holographic HUD, Motion posters, animating texts, shapes and masking, trim path, and different type of effects, Path Animation, Puppet Tool,	**15**
III	**Introduction Effects** Parallax animation, understanding assets and effects, Chroma key, Tracking and Stabilizing and Mocha Tracking	**15**
IV	**Project** Students will have to individually submit project of each sub contents in a storage device. Teacher will supervise the projects	**15**
	TOTAL	**60**

Textbook:

1. Brown, Blain; Cinematography: Theory and Practice, Second Edition: Image Making for Cinematographers and Directors; Focal Press, 2011.
2. Katz, D Steven; Film Directing Shot by Shot: Visualizing from Concept to Screen; Michael Wiese

References:

3. https://www.youtube.com/watch?v=KRC-6fMbw7k
4. https://www.youtube.com/watch?v=bVlTlTRhRDc

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	30 hrs. Project (5hrs),Study tour(4hrs), shooting (20hrs),

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time(hrs.)
Project	NA	5	5
Study Tour	NA	4	4
Shooting	NA	20	5
Interaction with industryexpert	NA	1	1
		Total Hours	**30**

Semester IV

Major Course: 1

Title of the Paper: 2D Animation FX and Compositing **Course Level: 100**

Subject Code: AVE092M411

L-T-P-D : 1-0-4-3 **Total credits: 4**

Course Objectives:

To define the concepts, techniques, principles and practices in Advertising and Public Relations in order to classify the mysteries of media marketing, positioning, market segmentation and targeting in advertising as well as the significance of media in globalization.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Review** 2D effects animation (also known as FX animation) special visual effects used in animation to create complex and dynamic effects or natural elements.	**BT 1**
CLO 2	**Picture graphically** each movement timing, spacing, and squash and stretch. These principles help to create a sense of weight, momentum, and fluidity in the animation	**BT 2**
CLO 3	**Illustrate** with of technical proficiency, and attention to detail to create smooth and realistic 2d animation that captures the essence of the movement trying to depict.	**BT 3**
CLO 4	**Break Down** Pseudo 3d (2.5D) and compile the shot	**BT 4**

COURSE OUTLINE:

Modules	**Course Contents**	**Periods**
I	**Advance 2D Animation** Separating biomechanics in performance, character dissection, rigging, tween classic | shape |motion animation and Lip-syncing with sound Character Coloring, Character Light and Shade, Guide Layer, Masking, introduction to Motion Path Animation and tweening, camera	**15**

II	**FX** Water, air, fire, sand and other various dynamics of 2D animation; Understanding physics in 2D; Pseudo 3d (2.5D)	**15**
III	**Compositing** Compositing 2D animation with background, Colour grade, Lighting and FX	**15**
IV	**Projects** Students will have to individually submit project of each sub content in a storage device. Teacher will supervise the projects.	**15**
TOTAL		**60**

Text Book:

1. Williams, Richard. (2001). *The Animator's Survival Kit*

References:

1. https://www.youtube.com/watch?v=fJvcllrOrr4
2. https://www.youtube.com/watch?v=XHprIlkY8Q4.

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Inspiration board / reference/visualization technique (6hrs), Project and Portfolio (15hrs), Conceptual Sketching (6hrs), Case studies (3hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)

inspiration board / reference/visualization technique	NA	6	6
Project and Portfolio	NA	15	15
Conceptual Sketching	NA	6	6
Case studies	NA	3	3
		Total Hours	**30**

Major Course: 2
Title of the Paper: Rigging & 3D Animation Techniques

Subject Code: AVE092M412 **Course Level: 100**

L-T-P-D : 1-0-4-3 **Total credits: 4**

Course Objectives:

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Identify** different type of editors that govern the movement and behavior of objects.	**BT 1**
CLO 2	**Understanding** the setup of different types of rigging for organic and inorganic models.	**BT 2**
CLO 3	**Experiment** with the form of animations that can be related with different type of materials, sizes, shapes and characters.	**BT 3**
CLO 4	**Corelate** the various elements of rigging with different types of animation using principles of animation.	**BT 4**

COURSE OUTLINE:

Modules	**Course Contents**	**Periods**
I	**Introduction to Nodes & Connections:** Node Editor, Utility Nodes, Connection Editor & SDK, Constraints, Joints and FK/IK Handles,	**15**
II	**Rigging & Skinning of organic and inorganic** Arm Set-up, Leg Set-up, setting up the Skeleton, Finishing the Rig with Controls, Skin Binding and Painting Weights, Facial Rigging, Character Pose with self-	**15**

	reference.	
III	**Animation Principles** Introduction to Animation (Timeline, Slider, Key Frames), Graph Editor, Different materials of bouncing balls together with Concept,	**15**
IV	**Animation Exercise** Biped progressive walk, Run Cycle, Jump with distance, Basic face expressions with Joy, anger, shock, etc.	
	TOTAL	**60**

Textbook:

1. **3D Animation Essentials: New Directions for Evaluation**
2. **3D Animation for the Raw Beginner Using Maya by R. King**

References:

1. https://youtube.com/playlist?list=PLoxdv8fALl90LfISWtg2GaOFmjttfCtE9&si=WrSZevzpBpsAblu2
2. https://youtube.com/playlist?list=PLnRp1fUkiXQf7VPf1TmVosgKItQBtdsq5&si=6BjuiHrSiymZHqef

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Conceptual Prototype (10hrs), Project and Portfolio (15hrs), (3hrs), Case studies (5hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Conceptual Prototype	NA	8	12
Project and Portfolio based on trends	NA	10	20
Case studies	NA	5	5
		Total Hours	**30**

Major Course: 3

Title of the Paper: Introduction to Chitra Art with Mograph (IKS based) **Course Level: 100**

Subject Code: AVE092M413

L-T-P-E : 1-0-4-3 **Total credits: 4**

Course Objectives:

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Studying** of traditional Indian art forms, such as Madhubani art, Warli art, or Pattachitra, can inspire modern animation techniques.	**BT 1**
CLO 2	**Relate** the use of visual metaphors, storytelling techniques, and character design to convey deeper cultural meanings.	**BT 2**
CLO 3	**Applying** the incorporation of these unique visual styles into contemporary animation.	**BT 3**
CLO 4	**Analyze** how Indian mythology, folklore, and traditional art forms can be effectively portrayed through animation.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Illustrator Basics** Overview of Illustrator Interface, Navigation and Workspace, Customization, Understanding Artboards, Using the Pen Tool, Creating Basic Shapes, Editing Paths and Anchor Points	**15**
II	**Organizing and Exporting Objects** Organizing Artwork with Layers, Creating and Editing Masks, Using Clipping Masks,	**15**
III	**Introduction to Mograph** Motion Paths and Graph Editor, Shape Layers and Vector Graphics, Complex Keyframing	**15**
IV	**Project** Students will have to submit a project of each sub-contents using only the predefined software.	**15**
	TOTAL	**60**

Textbook:

- Madhubani Art: Indian Art Series by Bharti Dayal
- Way: An Easy Introduction to Adobe Illustrator

References:

1. https://www.youtube.com/watch?v=2AqV-UzHxn8
2. https://www.youtube.com/watch?v=Ib8UBwu3yGA

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	30 hrs. Project (5hrs), Study tour(4hrs), shooting (20hrs),

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time(hrs.)
Project	NA	5	5
Study Tour	NA	4	4
Shooting	NA	20	5
Interaction with industryexpert	NA	1	1
		Total Hours	**30**

Minor Course: 1

Title of the Paper: Basics of Video Editing **Course Level: 100**

Subject Code: AVE092N411

L-T-P-F : 1-0-4-3 **Total credits: 3**

Course Objectives:

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Recognize** the basics of camera, how it works, and its functions. Distinguish between various camera body parts, different shot sizes and various camera setups using Rule of thirds for better composition.	**BT 1**
CLO 2	**Understanding** the fundamentals of lighting that affects the visual mood and atmosphere of a scene. It also helps in better storytelling, shaping character experience and enhancing the cinematic lighting techniques.	**BT 2**
CLO 3	**Apply** the knowledge of software to construct a product from different shots.	**BT 3**
CLO 4	**Analyze** the final product/composition containing camera, lighting, designing characters and story boarding/scripting to explore the new dimensions of audio video editing.	**BT 4**

COURSE OUTLINE:

Modules	**Course Contents**	**Periods**
I	**Fundamentals of Camera** Basics of Camera, its functions, Single camera and multi-camera setup, accessories, composition, Rule of thirds, different types of shots.	**15**

II	**<u>Introduction of Lighting</u>** Importance of lighting,3 point Lighting, 5 point Lighting,	**15**
III	**<u>Video Software</u>** Adobe Premiere Introduction, hardware requirements; capturing; Timeline in depth; mixing; Exporting – all the video formats.	**15**
IV	**<u>Video project</u>** Students will have to shoot and edit a short documentary/film and submit for the fulfilment of the course. The film will be scripted, shot and edited by individual/group student for the fulfilment of the course.	**15**
TOTAL		**60**

Textbook:

1. Brown, Blain; Cinematography: Theory and Practice, Second Edition: Image Making for Cinematographers and Directors; Focal Press, 2011.
2. Katz, D Steven; Film Directing Shot by Shot: Visualizing from Concept to Screen; Michael Wiese

References:

1. https://www.youtube.com/watch?v=KRC-6fMbw7k
2. https://www.youtube.com/watch?v=bVlTlTRhRDc

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	<u>30 hrs.</u> Project (5hrs),Study tour(4hrs), shooting (20hrs),

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time(hrs.)
Project	NA	5	5
Study Tour	NA	4	4
Shooting	NA	20	5
Interaction with industryexpert	NA	1	1
		Total Hours	**30**

Minor Course: 2

Title of the Paper: Fundamentals of Visual Effects **Course Level: 100**

Subject Code: AVE092N412

L-T-P-G : 1-0-4-3 **Total credits: 3**

Course Objectives: This course is designed to introduce students to Adobe After Effects and its core functions, focusing on composition creation, visual effects, and motion graphics. Students will learn to navigate the After Effects interface, understand the essential tools for animation, and gain proficiency in rendering techniques. The course will also cover key visual effects techniques, such as path animation, puppet tool animation, camera projection, and the basics of tracking, stabilizing, and green screen removal. Additionally, students will explore advanced animation techniques for texts, shapes, and masking. The course will culminate in individual projects that demonstrate the practical application of these techniques.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Identify** the holographic sequence and overall flow of the story and whether the images effectively convey the intended narrative.	**BT 1**
CLO 2	**Relate** the live action world with VFX to transform it into Imaginary world using VFX software	**BT 2**
CLO 3	**Apply** the types of typography of different text to be able to adapt for creating any graphics.	**BT 3**
CLO 4	**Analyze** the knowledge of VFX to create appealing composition.	**BT 4**

COURSE OUTLINE:

Modules	**Course Contents**	**Periods**
I	**Introduction to After Effects** Introduction to the UI of software, composition, tools, encoder, and rendering	**15**

II	**Fundamentals of Visual Effects-** Path Animation, Puppet tool Animation, Camera Projection, Understanding assets and effects and Stabilizing, Tracking, Green Screen Removal	**15**
III	**Visual Effects-** Animating texts, shapes Animation, masking, trim path,	**15**
IV	**Video project-** Students will have to individually submit project of each sub contents in a storage device. Teacher will supervise the projects	**15**
TOTAL		**60**

Textbook:

1. Adobe After Effects Classroom in a Book
2. Hands-On Motion Graphics with Adobe After Effects

References:

3. https://www.youtube.com/watch?v=Ar7KMoY4lE0
4. https://www.youtube.com/watch?v=hb2bbfiNBXA

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	30 hrs. Project (5hrs),Study tour(4hrs), shooting (20hrs),

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time(hrs.)
Project	NA	5	5
Study Tour	NA	4	4
Shooting	NA	20	5
Interaction with industryexpert	NA	1	1
		Total Hours	**30**

5^{th} SEMESTER

Major Course: 1

Title of the Paper: Compositing & CGI Integration **Course Level: 300**

Subject Code: AVE092M511

L-T-P-C: 1-0-4-3 **Total credits: 4**

Course Objectives: This course aims to equip students with essential and advanced skills in digital compositing, integrating CGI elements, and enhancing visual storytelling through seamless post-production techniques. Students will gain hands-on experience with industry-standard software such as Adobe After Effects and The Foundry Nuke, learning to composite live-action footage with computer-generated imagery and visual effects.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the fundamentals of digital compositing, including techniques like matte painting, chroma keying, and rotoscoping**.**	**BT 1**
CO 2	**Demonstrate** the process of integrating CGI elements into live footage using tracking, camera projection, and 3D elements.	**BT 2**
CO 3	**Apply** advanced compositing techniques using software such as AfterEffects and Nuke, incorporating color grading and optical	**BT 3**
CO 4	**Classify** a complete compositing project showcasing learned techniques, demonstrating creativity, technical ability, and visual	**BT 4**

COURSE OUTLINE:

Modules	**Course Contents**	**Periods**
I	**Fundamentals of Compositing:** Digital Compositing, Camera Projection and Animation, ***Understanding Mattes and Chroma Keying:*** Green Screen & Chroma Keying, Rotoscoping, Matte Painting, Sky Replacement	**15**
II	**CGI Integration & Advanced Compositing:** Tracking, Camera Tracking & Match moving, Motion Tracking (2D & Mocha Integration), Working with 3D Elements in After Effects, Color Correction & Grading, working with 3D Plugins Element 3D, Enhancing CGI with Optical Effects (Lens Flares, Glows, Distortion)	**15**
III	**Introduction to The Foundry Nuke:** Introduction to the UI of the software, Composition, Tools, Encoder and Rendering; Understanding node system; Image projection, using effects and node system to add visual effects, Color correcting footage.	**15**
IV	**Project:** Students will have to individually submit project of each sub content in a storage device. Teachers will supervise the projects.	**15**

Textbook:

- Nuke 101: Professional Compositing and Visual Effects by Ron Ganbar, published by Peachpit Press.

  https://ptgmedia.pearsoncmg.com/images/9780321733474/samplepages/0321733479.pdf

- Adobe After Effects Official Documentation (https://helpx.adobe.com/after-effects/user-guide.html)
- Foundry Nuke Official Documentation (https://learn.foundry.com/nuke)
- The Art and Science of Digital Compositing by Ron Brinkmann, published by Morgan Kaufmann.
- Matchmoving: The Invisible Art of Camera Tracking by Tim Dobbert, published by Sybex.

Major Course: 2

Title of the Paper: Overview of Dynamics **Course Level: 300**

Subject Code: AVE092M512

L-T-P-C: 1-1-4-4 **Total credits: 4**

Course Objective: This course is designed to provide students with in-depth knowledge and practical skills in soft body dynamics, hair simulation, n-Cloth, and particles within a 3D animation software, focusing on the simulation and rendering of realistic physical behavior and interactions. By the end of the course, students will apply these techniques to individual projects under supervision.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify** the fundamentals of soft body, rigid body, n-cloth, dynamic constraints, fields, and solvers.	**BT 1**
CO 2	**Relate** Maya's native dynamics to simulate collisions, destruction effects, and secondary motion.	**BT 2**
CO 3	**Apply** n-Cloth properties to simulate cloth behavior on different surfaces and objects.	**BT 3**
CO 4	**Examine** the application of various dynamic simulations by showcasing a complete dynamics-based project.	**BT 4**

Detailed Syllabus:

Modules	Course Contents	Periods
1	Introduction to Soft body, Rigid body, n-cloth, Dynamic Constraints, Fields & Solvers.	15
2	Maya's native dynamics, simulating **collisions** (falling objects, destruction effects) Adding secondary motion like **bouncing, rolling, stacking**	15
3	Playing with n-Cloth properties, Applying cloth on human body, and Flag simulation with n-cloth.	15
4	**Project** Students will have to individually submit project of each sub contents in a storage device. Teacher will supervise the projects	15
	TOTAL	**60**

Books:

- **"Learning Maya 6: Dynamics"** – A comprehensive guide to mastering Maya's dynamics tools, covering everything from particle systems to rendering
- **"Maya Studio Projects: Dynamics" by Todd Palamar –** This book provides hands-on projects to create realistic simulations like fire, water, and tornadoes.

References:

- https://help.autodesk.com/view/MAYAUL/2024/ENU/?guid=GUID-E1498F66-BD9D-4DB9-9BB7-EA123ABEB9E7

DSE: 1

Title of the Paper: 2D Rigging and Animation **Course Level: 300 Subject**

Code: AVE092D511 **Total credits: 4**
L-T-P-C: 1-1-4-4

Course Objective: Students will gain a comprehensive understanding of the animation production process, from the initial planning stages to the final output. They will develop practical skills in creating hand and expression sheets, model sheets, background sheets, and storyboarding. The course will cover the importance of keyframe animation, in-betweening, lip-syncing, and coloring, as well as technical skills such as character rigging and the creation of animatics.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Review** storyboard and animatic *sequence* and overall flow of the story and whether the images effectively convey the intended narrative.	**BT 1**
CO 2	**Picture graphically** each movement timing, spacing, and squash and stretch to create a sense of weight, momentum, and fluidity in the animation	**BT 2**
CO 3	**Illustrate** with technical proficiency, and attention to detail to create smooth and realistic 2D animation that captures the essence of the movement trying to depict.	**BT 3**
CO4	**Demonstrate the knowledge of Rigging and animation**	**BT 4**

Detailed Syllabus:

Modules	**Course Contents**	**Periods**
1	**Industrial Pipeline:** Plan animation production and set deadline for production	15

2	**Rigging:** Hand and expression sheet, eye blink, Model sheet, background sheet, and storyboarding, Layout process, X sheet/ dope sheet, scratch sound recording. Animatics, Character dissection and rigging tool	15
3	**Animation** Keyframe animation, in-between animation, test, clean up, test, lip-sync, Inking and Coloring	15
4	**Project and Portfolio** Student will submit the Project and portfolio	15
	TOTAL	**60**

References:

- The Animation Book: A Complete Guide to Animated Filmmaking

Books:

- **Stop Motion: Craft Skills for Model Animation**" by Susannah Shaw
- **The Advanced Art of Stop-Motion Animation"** by Ken A. Priebe

Minor: 1

Title of the Paper: Stop Motion **Course Level: 300**
Subject Code: AVE092N511 **Total credits: 4**
L-T-P-C: 1-1-4-4

Course Objective: student shall be guided on to execute Stop Motion Animation. Student shall cover every step in details, ranging from what and how a story for stop motion animation should be and how does it differ from other forms of storytelling

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** interface and workflow of the software for effective work	**BT 1**
CO 2	**Review** the storyboard overall flow of the story and whether the images effectively convey the intended narrative.	**BT 2**
CO 3	**Apply** the knowledge of color, shapes and forms to retain mood and the mass of subject to create smooth *sequence.*	**BT 3**
CO 4	**Demonstrate** the knowledge of stop motion	**BT 4**

Detailed Syllabus:

Modules	Name	Course Contents	Periods
I	**Introduction to Software and Staging**	Interface, Tool, Timeline, Properties, setup and shooting	**15**
II	**Storyboard**	Storytelling, Script, shot division /screenplay, storyboard	**15**
III	**Puppetry in stop motion**	Bio-Mechanic, 12 principles of animation, Time and spacing, arcs, multiple objects and replacement.	**15**
IV	**Project and portfolio**	Video reel with dialogue and Time	**15**
		Total	**60**

Textbook:

- Effects Stop Motion Animation: How to Make and Share Creative Videos
- Stop Motion: Craft Skills for Model Animation

References:

- 01 First Steps - Stop Motion Studio Tutorial
- Q4 IN Regional Pricing Update- CCI

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	30 hrs. Project (5hrs), Study tour(4hrs), shooting (20hrs),

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time(hrs.)
Project	NA	5	5
Stage set up	NA	4	4
Shooting	NA	15	10
Interaction with industryexpert	NA	1	1
		Total Hours	**30**

6^{th} Semester

Major Course: 1

Title of the Paper: Post-production for 2D Animation **Course Level: 300**

Subject Code: AVE092M611

L-T-P-E : 1-1-4-4 **Total credits: 4**

Course Objectives: This course aims to provide students with a thorough understanding of the principles and techniques required for creating basics of animation. It will cover the process of reviewing storyboards and animatics, converting them into fully-fledged 2D animations, and applying key animation and design principles, such as color, shapes, biomechanics, and fluidity. By the end of the course, students will be able to create biomechanics of animated sequences that effectively communicate a narrative while maintaining visual consistency.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Study** film noir through the visual style and aesthetic of a 2D animation film, which can be achieved through a combination of elements such as cinematic look, lighting, art style and colour grading.	**BT 1**
CLO 2	**Factors** that contribute to the way in which a 2D scene is captured on film or digital media are shot composition, and layout process, and they all contribute to the overall interpretation of the scene.	**BT 2**
CLO 3	**Apply** the knowledge of post-production to **compose** 2D animation shots.	**BT 3**
CLO 4	**Develop** new interpretations of contemporary ideas of animation based on an understanding of 2D animation production.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Industrial Pipeline on Pre-Production:** outline and brief, Script and Style, Storyboarding, Animation and Production, Audio Mix and Delivery	**15**

II	**Compositing in post-production:** Layering, Interpolation, Timeline Management, Fine tuning, Rendering	**15**
III	**Final Editing in post-production:** 2D Footage Editing	**15**
IV	**Project and Portfolio:** Student will submit the Project and portfolio	**15**
TOTAL		**60**

Textbook:

- Williams, Richard. (2001). *The Animator's Survival Kit*
- *Timing for Animation, 40th Anniversary Edition (greyscale)*

References:

- https://www.youtube.com/watch?v=fJvcllrOrr4
- https://www.youtube.com/watch?v=XHprIlkY8Q4.

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	30 hrs. Inspiration board / reference/visualization technique (6hrs), Project and Portfolio (15hrs), Conceptual Sketching (6hrs), Case studies (3hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)

inspiration board / reference/visualization technique	NA	6	6
Project and Portfolio	NA	15	15
Conceptual Sketching	NA	6	6
Case studies	NA	3	3
		Total Hours	**30**

Major Course: 1

Title of the Paper: Advanced 3D Dynamics **Course Level: 300**

Subject Code: AVE092M612

L-T-P-F : 1-1-4-4 **Total credits: 4**

Course Objective: By the end of this course, students will have developed a strong understanding and practical skills in the application of particle systems, hair styling, dynamics, and rendering in 3D environments.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Identify** and describe the fundamental concepts of particle systems, including menus, attributes, and emitters.	**BT 1**
CLO 2	**Explain** the process of creating and customizing different particle effects such as fire, smoke, explosions, and particles into glass.	**BT 2**
CLO 3	**Apply** techniques for hair styling, hair dynamics, and rendering, integrating these into character designs.	**BT 3**
CLO 4	**Analyze** and develop individual creative projects demonstrating the application of particle and hair system techniques.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	Particle System in 3D Dynamics: Introduction to particle Menus, attributes, Emitters Attributes.	**15**
II	Simulations: Particle Instancer, Creating fire, smoke, explosion, Particles into Glass.	**15**
III	Grooming: Creating hair on a character, Hair Styling, Hair Dynamics, and Rendering,	**15**
IV	Project Students will have to individually submit project of each sub contents in a storage device. Teachers will supervise the projects.	**15**
	TOTAL	**60**

Textbook:

- Hair: The Fascinating World of Hair, Its Growth, and Care" by Joseph J. M
- The Particle Adventure" by Eric T. L. and Ian M

References:

- https://www.youtube.com/watch?v=XHprIlkY8Q4.
- https://www.youtube.com/watch?v=UN9hxyg9Y-c

	Credit Distribution	
Lecture /Tutorial	**Practicum**	**Experiential Learning**
	60 hrs.	30 hrs. Inspiration board / reference/visualizationtechnique (6hrs), Project and Portfolio (15hrs), Conceptual Sketching (6hrs), Case studies(3hrs)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
inspiration board / reference/visualization technique	NA	6	6
Project and Portfolio	NA	15	15
Conceptual Sketching	NA	6	6
Case studies	NA	3	3
Total Hours			**30**

Discipline Specific Elective: (DSE) 1

Title of the Paper: Procedural Animation and Compositing for Film **Course Level: 300**

Subject Code: AVE092D611

L-T-P-G : 1-1-4-4 **Total credits: 4**

Course Objectives: This course equips students with essential advanced skills in compositing and VFX using Nuke and Houdini, focusing on seamlessly integrating live-action footage, CGI, and digital effects. Students will learn Nuke's node-based workflow for keying, tracking, rotoscoping, 3D compositing, and multi-pass integration, alongside Houdini's procedural tools for creating simulations like fire, water, and smoke. By the end, students will be able to combine these techniques to produce professional-quality VFX shots ready for industry pipelines.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Recall** the fundamental concepts of compositing, including Nuke interface, node system, image formats, rotoscoping, tracking, and rendering.	**BT 1**
CO 2	**Explain** intermediate compositing techniques such as keying, advanced roto, matte painting integration, multi-pass compositing, and color grading.	**BT 2**
CO 3	**Apply** advanced compositing and CGI integration techniques using Nuke and Houdini, including 3D camera projection, deep compositing, particle systems, and VFX integration.	**BT 3**
CO 4	**Analyze** a complete VFX shot using professional compositing workflows, combining 3D elements, live footage, and procedural effects into a seamless final composition.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Nuke & Compositing Basics** • Overview of Nuke Interface & Workflow • Understanding Nodes • Image Formats & Color Spaces • Basic Compositing Techniques (Transform, Merge, Keymix) • Working with Rotoscoping, Wire Removal, & Masking • Sky Replacement • Day to Night • Tracking Basics (2D Tracking) • Render and Output Settings	**15**
II	**Advanced Compositing Techniques** • Keying & Green Screen Removal (Keylight, Primatte, IBK Gizmo) • Advanced Rotoscoping Techniques • 3D Tracking & Camera Projection • Matte Painting Integration • Advanced Color Correction & Grading with Day and Night transition • Motion Blur & Depth of Field • Multi-Pass Compositing & AOVs	**15**
III	**3D in Nuke & VFX Integration** • Working with 3D Geometry & Cameras • Lighting & Shadows in Nuke • Particle Systems & Atmospheric Effects • Deep Compositing Workflow • Integration with 3D Software (Houdini, Maya) • Final Project: Full Shot Compositing	**15**

IV	**Introduction to Houdini & Procedural Effects** • Houdini Interface & Procedural Workflow • Understanding Nodes & Attributes • Creating Procedural Geometry • Introduction to VFX & Expressions • Particle Simulations & Dynamics • Pyro & Flip Fluids (Fire, Smoke, Water) • Exporting Simulations to Nuke for Compositing	**15**
	TOTAL	**60**

Textbook:

- **Advanced Animation and Rendering Techniques: Theory and Practice** - Alan Watt, Mark Watt
- **Compositing Visual Effects: Essentials for the Aspiring Artist** (2nd Edition) - Steve Wright

References:

- **The VES Handbook of Visual Effects (3rd Edition)** - Jeffrey A. Okun, Susan Zwerman

	Credit Distribution	
Lecture /Tutorial	**Practicum**	**Experiential Learning**
	60 hrs.	30 hrs. Inspiration board / reference/visualizationtechnique (6hrs), Project and Portfolio (15hrs), Conceptual Sketching (6hrs), Case studies(3hrs)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
inspiration board / reference/visualization technique	NA	6	6
Project and Portfolio	NA	15	15
Conceptual Sketching	NA	6	6
Case studies	NA	3	3
Total Hours			**30**

Discipline Specific Elective: (DSE) 2

Title of the Paper: Film Appreciation & Analysis **Course Level: 100**

Subject Code: AVE092M311

L-T-P-H : 1-1-4-4 **Total credits: 4**

Course Objectives: This course aims to provide students with a comprehensive understanding of film language, visual storytelling techniques, and the evolution of cinema and visual effects. Through the study of narrative structures, cinematography, mise-en-scène, editing, sound design, and film genres, students will develop the ability to analyze and appreciate films critically. Additionally, students will explore contemporary trends in animation and digital filmmaking, with a focus on the impact of VFX, virtual production, and AI technologies in modern storytelling.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Relate** to the history and evolution of the language of films and visual storytelling.	**BT 1**
CLO 2	**Explain** the key elements of filmmaking – narrative, cinematography, mise-en-scene, editing, sound, and genres.	**BT 2**
CLO 3	**Develop** the understanding of film analysis techniques & criticism	**BT 3**
CLO 4	**Examine** the contemporary trends in fimmaking	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Film Language & Visual Storytelling** • Understanding Film as an Art Form • Overview of the history of evolution of films • Overview of the history and evolution of Visual effects in Media	**13**

II	**Analyzing the Elements of Film** Narrative, Cinematography, mise-en-scene, editing, sound, Film genres	**13**
III	**Film Analysis Techniques & Criticism** • Film Criticism, Observation & Interpretation • Acting Techniques & Character Development • Symbolism, Metaphor & Subtext in Film Language	**13**
IV	**Contemporary Trends in Animation Filmmaking** • Evolution of Digital Filmmaking & Streaming Era • Influence of Technology (VFX, Animation, Virtual Production, AI in Filmmaking) • Practical: Analyzing a Short Film/Scene in Class	**13**
TOTAL		**52**

Text Book:

1. **Film Art: An Introduction** by David Bordwell & Kristin Thompson
2. **A Short Guide to Writing About Film** by Timothy Corrigan
3. **Understanding Movies** by Louis Giannetti

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
	60 hrs.	<u>30 hrs.</u> Conceptual Prototype (10hrs), Project and Portfolio (15hrs), (3hrs), Case studies (5hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Conceptual Prototype	NA	8	12
Project and Portfolio based on trends	NA	10	20
Case studies	NA	5	5
		Total Hours	**30**

Minor Papers: 1

Title of the Paper: Pencil to Pixel **Course Level: 300**

Subject Code: AVE092N611

L-T-P-H : 1-1-4-4 **Total credits: 4**

Course Objectives: Student shall execute 2d Animation. Student shall cover every step in details, ranging from what and how a story for stop motion animation should be and how does it differ from other forms of storytelling.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Study** animation skills in using industry-standard animation software	**BT 1**
CLO 2	**Review** the storyboard overall flow of the story and whether the images effectively convey the intended narrative.	**BT 2**
CLO 3	**Apply** the knowledge of color, shapes and forms to retain mood and the mass of subject to create smooth *sequence.*	**BT 3**
CLO 4	**Demonstrate** the knowledge of Bio-Mechanics and **Construct** animated sequence	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to the software:** Interface, Tool, Timeline, Properties, tweening	15
II	**Digital animation** 12 principles of digital animation	15
III	**Storyboard and character** Storytelling, Script, shot division screenplay, storyboard/ animatic and simple shapes, character building and rigging	15
IV	**Project** Scene to animate	15
TOTAL		**60**

Textbook:

- How to Make Animated Films: Tony White's Complete Masterclass on the Traditional Principals of Animation Stop Motion: Craft Skills for Model Animation
- Adobe Animate CC Classroom in a Book

References:

- Evergreen_16x9_ProjectManagement_UGC_20s_la.EN Q4 IN Regional Pricing Update- CCI

NOTIONAL CREDIT HOURS (NCH)DISTRIBUTION (1C = 30 hrs, 3x30=90		
Lecture /Tutorial	Practicum	Experiential Learning
20hrs	40 hrs.	30 hrs. Project (5hrs), Study tour(4hrs), shooting (20hrs),

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Project	NA	5	5
Study Tour	NA	4	4
Shooting	NA	20	5
Interaction with industryexpert	NA	1	1
Total Hours			**30**

7th Semester

Major Course: 1
Subject: Concept for Game Design and Prototype **Course Level: 400**

Subject Code:

L-T-P-C : **1-1-4-4** **Total credits: 4**

Course Objectives students construct their own games with visual storytelling, game mechanics and strategic gameplay. Devise rules that engage players and prototype components like cards, tokens and boards. Through hands-on playtesting and critique, refine your ideas into playable games that challenge and entertain.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the basics of Game Play, Game theory, Equilibrium	**BT 1**
CO 2	**Review** the interface of the software and explore Nash Equilibrium	**BT 2**
CO 3	**Apply** knowledge of Game Design to build the blueprint	**BT 3**
CO 4	**Constructing** game prototype	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Game:** Game players, strategy, payoff, information set, equilibrium	15
II	**Game Theory Strategies and Equilibrium:** Types of game theory, Strategies, Nash Equilibrium, Software uses	15
III	**Game Design:** Types, game genres, game moment and RPGs, Blueprint, Game environment, game component	15
IV	**Project Prototyping** of analog, Digital, gamification	15
	TOTAL	60

Detailed Syllabus:

Modules	Course Contents	Periods
I	**Game theory:** What is Game, players, strategy, payoff, information set, equilibrium, Impact of game theory? Types of game theory	15
II	**Game Theory Strategies and Equilibrium:** Types of Game Theory Strategies, Nash Equilibrium, Software	15
III	**Game Design:** Game types, game genres, game moment and RPGs, Blueprint, game component	15
IV	**Project:** Prototyping of analog, Digital, gamification	15
	TOTAL	60

Textbook:

- The Book of Board Games: An Introduction to Modern Tabletop Gaming
- Make Your Own Board Game: Designing, Building, and Playing an Original Tabletop Game

References:

- Introduction to (Tabletop) Role-playing Game Design | Udemy

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation(hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Prototype making	NA	10 hrs	10 hrs
Study Tour	3 hrs	1 hrs	4 hrs
Brainstorming	4 hrs	1 hrs	5 hrs
Interacting with expert	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs
Total Hours			**30**

Major

Course: 2
Subject: Research Methodology **Course Level: 400**

Subject Code:

L-T-P-C : **1-1-4-4** **Total credits: 4**

Course Objectives: Students would be able to think critically, creatively and independently, To critically evaluate in-depth information from diverse sources, To do research work and write research report independently

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	To make the students aware about research culture among academics and professionals in different fields	**BT 1**
CLO 2	To undertake research in their specific academic fields on philosophical, epistemological understanding of the elements of research	**BT 2**
CLO 3	To provide knowledge skills of various types of research designs and its procedures to conduct research to meet the national and international requirements	**BT 3**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Research** Definition & Concept (Variables, Hypothesis, Theory), Objectives, scope and importance of Research, Types of research, Approaches to research (quantitative and qualitative and mixed method); Steps in research, Concept of reliability, Validity.	**13**
II	**Methods & methodologies of Research** Qualitative- Quantitative Technique, Content Analysis, Sample Survey Method, Observation Methods, Experimental Studies, Case Studies, Interview Method, Focus Groups, Ethnography Narrative Analysis, Historical research, Sampling techniques.	**13**
III	**Data Collection** Data Collection Techniques, Primary Data, Secondary Data, Methods of collecting data, Statistical Analysis (Descriptive & Inferential Statistics), Central Tendency, Dispersion, Presentation, and Interpretation of Research Findings	**13**
IV	**Report Writing** Research Proposal and Report Writing, Presentation of research reports, Referencing and Citation Style, Ethics in Research	**13**

	TOTAL	52

Text Books:

- *Mass Media Research*, Roger, Wimmer. D and Dominick, Joseph,R; Thomson Wadsworth; 2006.

Reference Books:

- Berger, Arthur Asa; *Media Research Techniques*; Second Edition; Sage Publications, New Delhi; 1998.
- Fiske, John; *Introduction to Communication Studies*; Third Edition; Routledge Publications; 1982.
- Croteau, David and Hoynes; *Media/Society: Industries, Images and Audiences;* William; Forge Press; 2002.
- Kothari, C.R; *Research Methodology: Methods and Techniques*; New Age International Ltd. Publishers; 2013.

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
R&D	NA	10 hrs	10 hrs
Field Study	3 hrs	1 hrs	4 hrs
Creative Writing	4 hrs	1 hrs	5 hrs
Interacting with expert	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs
		Total Hours	**30**

Major Course: 3
Subject: Overview of UI/UX **Course Level: 400**

Subject Code:

L-T-P-C : 1-1-4-4 **Total credits: 4**

Course Objectives:

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Remember** the principles of interface design and the different methods undertaken to study user experiences.	**BT 1**
CLO 2	**Relate** user-centered design principles to the creation of effective UI/UX solutions.	**BT 2**
CLO 3	**Apply** wireframing, prototyping, and usability testing techniques to develop functional design outputs.	**BT 3**
CLO 4	**Analyze** the designs to plan and execute a final quality output of the prototype.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to User Interface**: UI history, User Centered Thinking, Context and Empathy, UI Design Process, User Xperience, Action History, UX processes and development	15
II	**Wire framing:** Design wireframes for a website or app based	15
III	**Prototyping:** Turn a wireframe into an interactive prototype, Usability Testing	15
IV	**Mobile Design:** Redesign an existing website with a mobile-first approach	15
	TOTAL	**60**

Books-

- **Don't Make Me Think: A Common Sense Approach to Web Usability"** by Steve Krug

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
R&D	NA	10 hrs	10 hrs
Studio Visit	3 hrs	1 hrs	4 hrs
Brainstorming	4 hrs	1 hrs	5 hrs
Interacting with UI/UX expert	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs
		Total Hours	**30**

Discipline Specific Elective (DSE): 1
Subject: Visual Effects Production **Course Level: 400**

Subject Code:

L-T-P-C : **1-1-4-4** **Total credits: 4**

Course Objectives: This course provides students with a complete understanding of the VFX production pipeline, combining technical skills, creative techniques, and industry knowledge. Through hands-on projects, students will gain experience in 3D modeling, animation, FX simulations, and compositing, along with essential pre-production skills like storyboarding and previs. Emphasis is placed on seamlessly integrating CG with live-action footage to achieve realistic effects. By the end of the course, students will create a professional showreel, preparing them for entry-level roles in film, TV, advertising, and games.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Define** the VFX production pipeline and demonstrate an understanding of pre-production processes such as storyboarding, cinematography, and previs.	**BT 1**
CLO 2	**Classify** 3D asset creation techniques, including modeling, texturing, shading, lighting, animation, and FX simulations, using industry-standard tools.	**BT 2**
CLO 3	**Apply** compositing workflows, including match moving, rotoscoping, multi-pass compositing, color grading, and integrating CG into live-action footage.	**BT 3**
CLO 4	**Analyze** a final VFX project, demonstrating project planning, asset management, rendering optimization, and development of a professional showreel tailored for the VFX industry.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Visual Effects & Pre-Production** 1. Understanding VFX & Industry Overview 2. VFX Production Pipeline 3. Concept Development & Storyboarding 4. Introduction to Camera & Cinematography for VFX 5. Introduction to Previsualization (Previs)	**15**
II	**3D Asset Creation & Animation** 1. 3D Modeling for VFX 2. Shading, Lighting & Rendering 3. Introduction to Animation & Rigging 4. Particle Effects & Simulations o Using Houdini / Blender for FX Simulations o Fire, Water, Smoke, Destruction FX o Procedural Workflows	**15**

III	**Compositing & post-production** 1. Introduction to Compositing 2. Matchmoving & Rotoscoping 3. Advanced Compositing Techniques o 3D Projection & Camera Mapping o Multi-Pass Compositing o Color Grading & Look Development 4. Integration of CG with Live-Action o Shadow Matching & Reflections o Depth & Parallax Effects o Advanced Matte Painting	**15**
IV	**Final Project & Industry Preparation** 1. Project Planning & Asset Management o Shot Breakdown for Production o Using ShotGrid or Production Tracking Tools 2. Rendering & Optimization o Render Layering Techniques o GPU vs CPU Rendering o Render Passes & Post-Processing 3. Portfolio Development & Showreel Creation o Choosing & Editing Best Shots o Creating a Demo Reel for VFX Studios	**15**
	TOTAL	**60**

Textbook

1. Producing VFX: A Guide to Managing the Process by Charles Finance & Susan Zwerman
2. Filming the Fantastic: A Guide to Visual Effects Cinematography by Mark Sawicki

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)
Break up of Experiential learning		

Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Extracurricular Activity	NA	10 hrs	10 hrs
Cultural activities	3 hrs	1 hrs	4 hrs
Street Play	4 hrs	1 hrs	5 hrs
Interacting with actor	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs
		Total Hours	**30**

Minor Course: 1

Subject: Introduction to Architecture Modelling **Course Level: 400**

Subject Code: **Total credits: 4**

L-T-P-C: 1-1-4-4

Course Objectives: The objective of this course is to equip students with the fundamental skills required for architectural modeling, texturing, and rendering using **3ds Max**. Students will learn how to create detailed and realistic architectural models from various architectural styles, apply different textures and materials, and produce high-quality renders using industry-standard render engines.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the different form of architecture to visualize the art form for better composition.	**BT 1**
CO 2	**Convert** different era architecture into digital form.	**BT 2**
CLO 3	**Apply** the knowledge of texture to beautify an architecture of different era.	**BT 3**
CO 4	**Construct** appealing renders using models, texture, and lights.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Modelling**: Introduction to 3ds Max to create different types of Prehistoric, Modern, Classical, Vernacular, Contemporary Architectures, Importance of architecture in different historical periods	**15**
II	**Types of Modifiers:** Extrude, Modifiers, Object Modifiers, Edit Modifiers, and World-Space Modifiers	**15**
III	**Materials & Texturing:** Baking of different Maps such as Diffuse Map, Metallic Map, Roughness Map, Transmission Map, Height Map, Displacement Map, and Sub Surface Scattering Map, Texturing software to finalize the look	**15**

IV	**Physically Based Rendering** Using Render Engine like Arnold, V-ray, Lumion to export the final Render	**15**
	TOTAL	**60**

Books:

- Architectural Model Building: Tools, Techniques & Materials 1st Edition by Roark T. Congdon
- A Practical Study in the Discipline of Architectural Model making by James Taylor-Foster

References:

- https://www.thoughtco.com/architecture-timeline-historic-periods-styles-175996

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Extracurricular Activity	NA	10 hrs	10 hrs
Cultural activities	3 hrs	1 hrs	4 hrs
Street Play	4 hrs	1 hrs	5 hrs
Interacting with actor	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs

	Total Hours	30

8th Semester

Major Course: 1
Subject: Media Entrepreneurship **Course Level: 400**

Subject Code:

L-T-P-C : **3-1-0-4** **Total credits: 4**

Course Objectives:

* To understand the fundamentals of entrepreneurship in the context of media
* To develop critical thinking and problem-solving skills to address challenges in media entrepreneurship
* To learn about the legal aspects of Media Business in India

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Define** and explain fundamental concepts, nature, scope, and types of entrepreneurships, with a focus on media entrepreneurship.	**BT 1**
CLO 2	**Describe** various entrepreneurship theories and frameworks, and explain their relevance in the context of media startups.	**BT 2**
CLO 3	**Apply** ideation techniques, market analysis tools, and business planning strategies to identify and evaluate entrepreneurial opportunities in the media sector.	**BT 3**
CLO 4	**Analyze** legal, ethical, and regulatory frameworks to assess their implications in the context of starting and running media enterprises.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Introduction to Entrepreneurship** Meaning, definition and concept of entrepreneurship, Nature and scope of media entrepreneurship, Historical overview and evolution of media startups, Entrepreneurs versus inventors, Challenges and risks in media entrepreneurship, Types of entrepreneurs: Clarence Danhof Classification, Arthur H. Cole Classification, Classification Based on Ownership, Classification	13

	Based on the Scale of the Enterprise	
II	**Theories of Entrepreneurship:** Theories: Economic, Classical, Neo Classical, Psychological, Personality Traits, Need for achievement, Sociological, Anthropological entrepreneurial, Opportunity based Entrepreneurship theory, Resource based Entrepreneur, Financial Capital/ Liquidity, Social Capital or Social Network Theory, Entrepreneurial Motivation – The Needs Framework, Manifest Needs Theory	13
III	**Opportunities** Startup Ecosystem in India, Market analysis and identifying niche audiences, Ideation techniques and opportunity recognition in media, Sources of new Idea, creative problem solving, opportunity recognition, product planning and development, creating a Business Plan, Startup India, Stand-up India, Make in India, Digital India, Ministry of Skill Development and their initiatives, NSDA, NSDC	13
IV	**Legal Aspects** Regulations to set up a new business, Legal issues in setting up the organization, patents, business methods patents, trademarks, copyrights, trade secrets, licensing, product safety and liability, insurance, contracts, Indian Contract Act, 1872, Sale of Goods Act, 1930, The Competition Act, 2002, Ethical dilemmas in media entrepreneurship, social responsibility and community engagement	13
	TOTAL	**52**

Text Books:

- Dahiya, S. (2023). Digital First: Entrepreneurial Journalism in India, OUP, UK
- Berger, A.A. (2018). Media and Communication Research Methods: An Introduction to Qualitative and Quantitative Approaches. SAGE Publications
- Dimmick, J., & Rothenbuhler, E.W. (2017). The Routledge Handbook of Media Industries. Routledge.
- Doyle, G. (2016). Understanding Media Economics. SAGE Publications

Reference Book:

- Ferrier, Michelle & Mays, Elizabeth. (2017). Media Innovation and Entrepreneurship. Rebus Community.
- Lamont, Ian. (2017). Lean Media: How to Focus Creativity, Streamline Production, and Create Media That Audiences Love. I30 Media Corporation.
- Kolb, Bonita. M. (2020). Entrepreneurship for the Creative and Cultural Industries. Routledge.

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Extracurricular Activity	NA	10 hrs	10 hrs
Cultural activities	3 hrs	1 hrs	4 hrs
Street Play	4 hrs	1 hrs	5 hrs
Interacting with actor	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs
		Total Hours	**30**

Major Course: 2
Subject: Papercut puppetry **Course Level: 400**

Subject Code:

L-T-P-C : **1-1-4-4** **Total credits: 4**

Course Objectives: students shall be guided on to execute Paper cut puppetry Animation. Student shall cover what and how a story for Paper puppetry animation should be.

Course Learning Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CLO 1	**Understand** interface and workflow of the software for effective work	**BT 1**
CLO 2	**Review** the storyboard overall flow of the story and whether the images effectively convey the intended narrative.	**BT 2**
CLO 3	**Apply** the knowledge of Paper cut rigs to create smooth *sequence.*	**BT 3**
CLO 4	**Demonstrate** the knowledge of Paper cut mechanism	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Software and Staging** Interface, Tool, Timeline, Properties, setup, and shooting	**15**
II	**Storyboard** Storytelling, Script, shot division /screenplay, storyboard	**15**
III	**Puppetry** Time and spacing, arcs, multiple object and replacement.	**15**
IV	**Final portfolio** Video reel with dialogue and Time	**15**
	TOTAL	**60**

	Credit Distribution

Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	<u>30 hrs.</u> Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time(hrs.)
Shoot	NA	10 hrs	10 hrs
Storyboard	3 hrs	1 hrs	4 hrs
Making paper elements	5 hrs	4 hrs	9 hrs
Interacting with actor	NA	1 hrs	1 hrs
Case studies	NA	6 hrs	6 hrs
		Total Hours	**30**

Major Course: 3

Subject: Short Film for 2D Animation **Course Level: 400**

Subject Code: AVE092D511

L-T-P-C : **1-1-4-4** **Total credits: 4**

Course Outcomes: Students will gain a comprehensive understanding of the animation production process, from the initial planning stages to the final output. They will develop practical skills in creating 2D animation. The course will cover the importance of understating teamwork and Industry pipeline.

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Picture Graphically** the storyboard and animatic *sequence* and overall flow of the story and whether the images effectively convey the intended narrative.	**BT 1**
CO 2	**Illustrate** with technical proficiency, and attention to detail to create smooth and realistic 2D animation that captures the essence of the movement trying to depict.	**BT 2**
CO 3	**Demonstrate** the knowledge of 2d animation to generate short film	**BT 3**
CO 4	**Construct** animated short Film	**BT 4**

COURSE OUTLINE:

Modules	**Course Contents**	**Periods**
I	**Industrial Pipeline** Plan for animation production and set deadlines for production	**15**
II	**Per Production and production** Script, Storyboard Animatic, character background, Sound Animation workflow and animation production	**15**
III	**Postproduction** Compositing and Editing	**15**
IV	**Final Project** Student will submit the Project (Short Film) duration 1 min	**15**
	TOTAL	**60**

Textbook:

- The Animation Book: A Complete Guide to Animated Filmmaking
- Animator's Survival Kit

References:

i. Making an Animated Movie course | 30 HD Video Lessons

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Production	NA	10 hrs	10 hrs
Scratch Sound	3 hrs	1 hrs	4 hrs
Animatics	1 hrs	4 hrs	5 hrs
Interacting with animator	NA	5 hrs	5 hrs
Case studies	NA	6 hrs	6 hrs
		Total Hours	**30**

Major Course: 4

Subject: 3D Portfolio Development **Course Level:** 300

Subject Code: **Total credits:** 4

L-T-P-C: 1-1-4-4

Course Objectives: To equip students with industry-standard 3D production pipeline knowledge and to develop a professional portfolio tailored to animation, gaming, VFX, or product visualization industries. To encourage creativity while adhering to global industry workflows and to enhance collaboration, communication, and presentation skills.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify and describe** the complete 3D production pipeline, industry roles, software, and documentation standards.	**BT 1**
CO 2	**Explain** the process of asset creation, optimization techniques, and technical checks used in industry workflows.	**BT 2**
CO 3	**Apply** industry-standard tools and techniques to create optimized 3D assets, work collaboratively on a simulated project, and manage asset pipelines.	**BT 3**
CO 4	**Analyze** portfolio work, incorporating peer and industry feedback, and present assets using professional standards.	**BT 4** BT 4 (Analyze)

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Industry pipeline and understanding-** Introduction to 3D Production Pipelines: Pre-production, Production & Post-production (Animation, Gaming, VFX & Product Design), Understanding Roles & Responsibilities (Modeler, Texture Artist, Animator, Lighting Artist, FX Artist, Compositor), Software, Tools (Industry-standard), Industry Documentation (Asset List, Naming Conventions, File Management), Analysing Showreels from Top 3D Studios & Artists	**15**
II	**Asset creation and optimization-** 3D Modelling for Portfolio (Hard Surface & Organic Modelling, High Poly vs Low Poly), Asset Optimization (for games, animation, or AR/VR), Texturing & UV Mapping (PBR Texturing Workflow, UV Layout Best Practices, Material Library Creation), Asset Integration (Importing to Game Engines/Rendering Software), Technical Checks (Polygon Count, Topology, Edge Flow), Asset Documentation (Turntable, Wireframe Renders, Texture Maps)	**15**

III	**Industry project simulation & collaboration- Simulated Industry Project (Teamwork)** (Assigning Roles (like a mini-studio), Developing Asset/Scene Pipeline)**, Project Management** (Using Trello/Shot-Grid, Milestone Tracking & Feedback Loops)**, Cross-platform Testing** (Portfolio-ready for Web, Unreal, Unity)**, Case Studies:** (Reviewing Successful 3D Portfolios from Industry Learning from Industry Failures)	15
IV	**Final Portfolio development & showcase-** Structuring the Portfolio (Asset Showcase, Breakdown Sheets, Wireframe, Textures, Final Renders), **Presentation Techniques** (Video Turntable, Marmoset Viewer, Sketch-fab)**, Personal Branding** (Online Portfolio Website (Art-Station, Behance), Resume & Showreel Integration)**, Peer Review & Industry Critique, Final Portfolio Submission**	15
	TOTAL	**60**

Textbook:

- "3D Art Essentials" by Ami Chopine
- "Digital Modelling" by William Vaughan
- "The Game Artist’s Guide to Maya" by Michael McKinley

Reference:

- CG Spectrum, Art -Station Learning, Gnomon, LinkedIn Learning
- Software Documentation (Maya, Blender, Substance Painter
- Autodesk Success Stories, GDC Portfolios

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning

Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Shoot	NA	10 hrs	10 hrs
Pre-Production	3 hrs	1 hrs	4 hrs
Interacting with animator	NA	1 hrs	5 hrs
Case Study	4 hrs	5 hrs	5 hrs
Post-Production	NA	6 hrs	6 hrs
		Total Hours	**30**

Major Course: 5

Subject: VFX – Portfolio Development **Course Level:** 400
Subject Code: **Total credits:** 4
L-T-P-C: 1-1-4-4

Course Objectives: This course prepares students to create a professional VFX portfolio and showreel aligned with industry standards. Students will analyze trends, identify target roles, curate their best work, and present their skills through effective storytelling and editing. Emphasizing technical refinement and career preparation, the course ensures students develop a strong online presence and gain essential tools like a resume, cover letter, and interview skills to confidently enter careers in VFX, film, gaming, and advertising.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI. No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the importance of VFX portfolios, analyse industry trends, and **identify** suitable roles and project types for the portfolio.	**BT 2**
CO 2	**Apply** asset collection, shot selection, and refinement techniques to curate high-quality content for a VFX showreel.	**BT 3**
CO 3	**Examine** the ability to edit, present, and **enhance** the showreel with storytelling, pacing, and professional formatting.	**BT 4**
CO 4	**Analyze** a complete portfolio package, including an online presence, resume, cover letter, and showreel, ready for industry submission.	**BT 4**

COURSE OUTLINE:

Modules	Course Contents	Periods
I	**Understanding Portfolio Requirements & Industry Standards** • Importance of a VFX Portfolio in the Industry • Analysing Showreels from Top VFX Studios & Artists • Understanding Target Roles (Compositor, FX Artist, Animator, Matte Painting Artist) • Choosing the Right Type of Projects for Your Portfolio • Researching VFX Industry Trends (Film, TV, Advertising, Games)	**15**
II	**Asset Collection & Shot Selection** • Reviewing Previous Coursework & Projects • Selecting Best Shots (Modelling, Animation, FX, Compositing) • Enhancing and Refining Existing Work	**15**
III	**Editing & Presentation Techniques** • Storytelling (Flow & Narrative) • Editing Techniques for Showreels (Timing, Pacing, Music Selection) • Adding Intro Titles, Shot Breakdowns, and Contact Information • Resolution, Formats, and Upload Guidelines (Vimeo, YouTube,	**15**

	Art-Station) • Receiving Peer & Instructor Feedback (Review Sessions)	
IV	**Final Portfolio Submission & Career Preparation** • Final Quality Check – Polishing Shots & Presentation • Submitting Final Portfolio for Evaluation • Creating an Online Portfolio/Website (Art-Station, Behance, Personal Site) • Writing an Effective Resume and Cover Letter for VFX Roles • Preparing for Portfolio Reviews & Interviews	**15**
	TOTAL	**60**

Textbook:

- VFX Artistry: A Visual Tour of How the Studios Create Their Magic by Spencer Drate
- The VES Handbook of Visual Effects (2nd Edition) by Jeffrey A. Okun & Susan Zwerman

	Credit Distribution	
Lecture /Tutorial	Practicum	Experiential Learning
NA	60 hrs.	30 hrs. Extracurricular Activity (10hrs), Cultural activities (preparation – 3hrs, presentation 1hrs), Street Play (1hrs on preparation, 4hrs on presentation), Interacting with actor (5hrs). Case studies (6hrs)

Break up of Experiential learning			
Activity	Time required for preparation (hrs.)	Time required for execution (hrs.)	Total Time (hrs.)
Shoot	NA	10 hrs	10 hrs
Pre-Production	3 hrs	1 hrs	4 hrs
Interacting with actor	NA	1 hrs	5 hrs

Case Study	4 hrs	5 hrs	5 hrs
Post-Production	NA	6 hrs	6 hrs
		Total Hours	**30**